# When Debiasing Hurts: An Exact Bias Identity for LLM-Annotated Treatment Effects

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## Abstract

Standard confusion-matrix correction for noisy annotations assumes uniform error rates. When this assumption fails—as it routinely does with large language model (LLM) classifiers, whose error rates vary by up to $5\times$ across subgroups—correction can reverse estimated effect signs. We derive an exact bias identity: global correction introduces subgroup-level bias $\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$ under symmetric errors (generalizing to a full-matrix form for arbitrary asymmetric confusion matrices; Remark 2), verified at $R^2 \geq 0.97$ on synthetic grids ($R^2 = 0.975$ in aggregate across 11 real LLM confusion matrices; per-model range 0.76–0.99). The identity predicts that global correction reverses conditional average treatment effect (CATE) sign in 14.9% of configurations where the true effect is small ($\tau(0) = 0.03$) and amplifies bias in all eight real LLMs tested; a $\Delta\tau$-free diagnostic corollary flags harmful correction from error rates alone, without estimating treatment effects. We characterize when per-subgroup correction is beneficial ($\Delta\pi > 0.15$, $n_{\text{expert}}/K \geq 50$, $K \leq 4$) versus when global correction should be retained, and implement one such correction via hierarchical Bayesian estimation, reducing |bias| by $7.4\times$ under extreme heterogeneity.

## 1 Introduction

Consider estimating whether a content moderation policy reduces hate speech differently across demographic groups, using GPT-4 to classify thousands of posts. After standard confusion-matrix correction, the analysis concludes the policy *increases* hate speech among younger users—but the true effect is protective. This sign reversal is not an edge case.

Large language models (LLMs) now routinely replace human annotators in computational social science, producing labels for sentiment, hate speech, stance, and political discourse at costs orders of magnitude below manual coding (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024; Tan et al., 2024). When these labels serve as effect modifiers in heterogeneous treatment effect (HTE) analysis, the standard correction protocol[1] is straightforward: validate a sample against expert annotations, estimate a confusion matrix, and apply its inverse uniformly to correct conditional average treatment effect (CATE) estimates. Yet LLM annotations are not uniformly accurate: Ashwin et al. (2025) document error rates varying by up to $5\times$ across demographic subgroups; in our pilot, GPT-4.1 error rates range from 10.2% to 20.0% across text-feature subgroups ($\Delta\pi = 0.098$).

We show that this standard correction can make things worse. In 14.9% of our synthetic configurations (across a grid of $\Delta\pi$ and $\tau(0)$ values; $\tau(0) = 0.03$), global correction reverses the sign of estimated CATE—a protective treatment ($\tau = +0.03$) appears harmful ($\hat{\tau} = -0.059$)—rising to 31% at $\tau(0) = 0.01$ (Figure 1); all eight real LLMs tested amplify bias under global correction, while our per-subgroup correction (EC-HTE; Section 4) eliminates sign reversal in the primary $K{=}2$ setting. Global correction amplifies bias by up to $5.9\times$ ($K{=}4$ with CausalForest); the classical Frazis–Loewenstein attenuation correction (Frazis and Loewenstein, 2003) *increases* bias by up to $2.9\times$. The mechanism is over-correction: pooled correction calibrated for the average error rate introduces substantial bias in low-error subgroups where the naive estimator is nearly unbiased.

We derive an exact bias identity explaining when and why (Theorem 1), verified at $R^2 \geq 0.97$ on synthetic grids. The per-subgroup correction prescribed by the identity reduces |bias| by $7.4\times$ under extreme heterogeneity; the advantage is conditional on sufficient $\Delta\pi$ and expert budget.

[1] Throughout, "debiasing" refers to correction of annotation measurement-error bias, not algorithmic fairness debiasing.

Existing corrections—prediction-powered inference (PPI) (Angelopoulos et al., 2023), design-based supervised learning (DSL) (Egami et al., 2023), prediction-powered conditional inference (PPCI) (Sui et al., 2026), arm-specific PPI (Mozer, 2026), ForestIV (Yang et al., 2023)—assume homogeneous prediction error; none addresses effect modifier misclassification (Section 2).

Our contributions are:

- An exact bias identity (Theorem 1): global correction introduces subgroup-level bias $\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$, explaining when and why it reverses CATE sign (14.9% of configurations at $\tau(0) = 0.03$); verified at $R^2 \geq 0.97$ on synthetic grids ($R^2 = 0.975$ in aggregate across 11 real LLM confusion matrices; per-model range 0.76–0.99). The identity immediately yields a $\Delta\tau$-free diagnostic (Proposition 5) predicting when correction increases bias from error rates alone, without estimating treatment effects (Sections 3–5.4).
- A "when not to correct" decision framework: per-subgroup correction outperforms global correction when $\Delta\pi > 0.15$, $n_{\text{expert}}/K \geq 50$, and $K \leq 4$; outside this envelope, the diagnostic guides practitioners to retain standard correction (Section 5.4).
- One concrete instantiation: hierarchical Bayesian confusion matrix estimation (EC-HTE), empirically verified for difference-in-means (DIM), CausalForestDML, and R-learner, and composable with PPCI for joint covariate/outcome error (Sections 4–5.4).

Code and data generation scripts are publicly available.[2]

## 2 Related Work

**Measurement error and annotation models.** Classical measurement error theory shows that correction via estimated error models can amplify mean squared error (MSE) when the error structure is misspecified (Fuller, 1987; Carroll et al., 2006); Bayesian approaches require informative priors when identification is weak (Gustafson, 2003), and nonparametric methods use repeated measurements or instruments to recover the true distribution (Hu and Schennach, 2008).

Three distinctions separate our contribution from prior measurement error analyses. First, we study effect modifier misclassification—the mismeasured variable defines the CATE function itself—rather than regressor (Frazis and Loewenstein, 2003; Carroll et al., 2006) or confounder (Ogburn and VanderWeele, 2012; Greenland, 1980) error, where attenuation is the primary concern. Second, Theorem 1 derives a bias identity for correction-induced bias—bias created by the debiasing procedure—rather than bounding the attenuation from misclassification itself. Concretely, Frazis and Loewenstein (2003) derive scalar attenuation $\lambda = (1 - 2\pi)$ for the uncorrected estimator under homogeneous misclassification; Theorem 1 yields a structurally interpretable decomposition—the product $\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$ whose sign varies across subgroups, revealing how error heterogeneity causes bias direction to reverse between subgroups, a phenomenon invisible to any global-scalar analysis. The homogeneous-error setting ($K{=}1$, $\delta_s{=}0$) is a degenerate case recovering classical attenuation (Blackwell et al., 2017); empirically, F-L correction produces higher bias under heterogeneous misclassification, while EC-HTE achieves near-oracle performance (Appendix Z.5). Third, the $\delta_s \cdot \Delta\tau$ product structure arises only under heterogeneous error rates; all homogeneous-error analyses (Battistin and Chesher, 2014; Fong and Tyler, 2021) miss this interaction entirely. In the annotation literature, Dawid and Skene (1979) introduced confusion matrix estimation for multiple annotators, with extensions to item difficulty, annotator competence, and hierarchical priors (Passonneau and Carpenter, 2014; Hovy et al., 2013; Paun et al., 2018). Northcutt et al. (2021) use per-class noise rates estimated from model confidence to identify mislabeled examples; our hierarchical estimator repurposes per-subgroup confusion matrix estimation for causal bias correction rather than label cleaning, adapting the Dawid–Skene architecture from label aggregation to causal inference (Section 4).

**HTE estimation and LLM annotation error.** Causal forests (Athey and Imbens, 2016; Wager and Athey, 2018), double/debiased ML (Chernozhukov et al., 2018; Semenova and Chernozhukov, 2021), doubly-robust CATE estimators (Kennedy, 2023), and meta-learners (Künzel et al., 2019) all assume that covariates defining heterogeneity are observed without error; when the effect modifier is misclassified, CATE estimates

[2] https://anonymous.4open.science/r/ec-hte

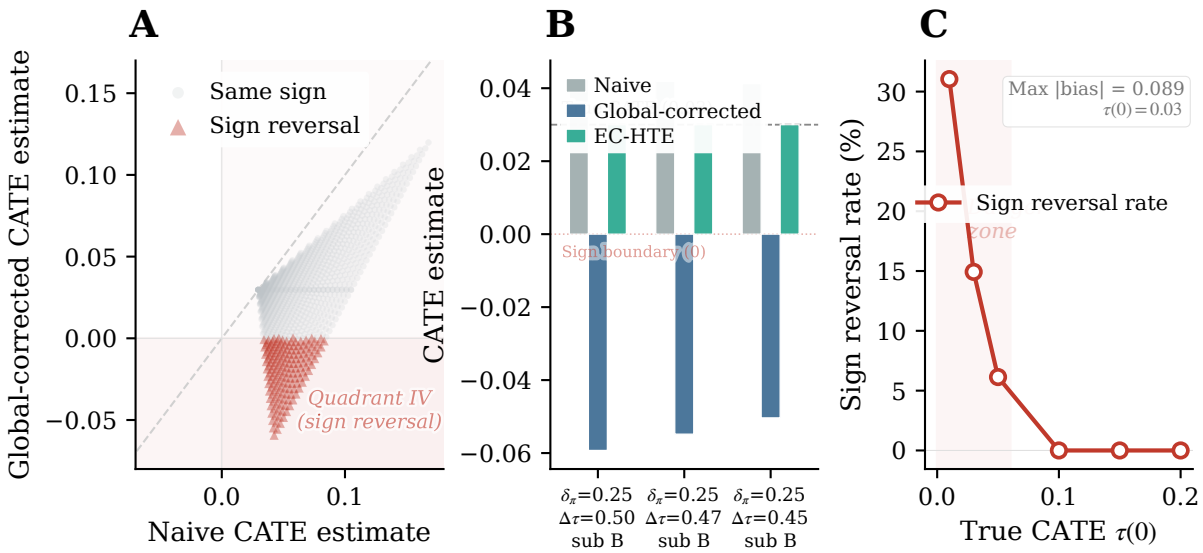



Figure 1: CATE sign reversal under global correction ($K$=2, $\tau(0) = 0.03$). (A) Naive vs. global-corrected CATE; red triangles mark sign reversals (7.5% of subgroup estimates). (B) Extreme case: true $\tau = +0.03$, estimated $\hat{\tau} = -0.059$. (C) 14.9% of configurations exhibit sign reversal; all 8 real LLMs amplify bias; EC-HTE: 0% reversal.

conflate effects across groups. Per-subgroup correction restores valid inputs to any downstream estimator. The corrected pseudo-outcomes are in principle compatible with other HTE estimators such as the R-learner (Nie and Wager, 2021). For broader perspectives on causal inference with text data, see Feder et al. (2022) and Keith et al. (2020); Wang et al. (2020) study inference correction when outcomes are ML-predicted, a related but distinct problem from effect modifier misclassification. LLMs now match crowd-worker accuracy on many classification tasks (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024; Pangakis et al., 2023), but Ashwin et al. (2025) show that error rates vary by up to 5× across demographic subgroups; Cheng et al. (2024) demonstrate that prompt optimization can reduce such error, a complementary preprocessing step. This subgroup-varying error distorts the very contrasts that define the research question; prior work documents the pattern without correcting downstream causal estimands. Concurrent work identifies variance-side LLM measurement error (Messing, 2026); Camuffo et al. (2025) characterize five variance sources in LLM annotation, complementary to our bias-side analysis. Gligorić et al. (2025) address valid inference with LLM annotations via confidence-driven data fusion, targeting population-level estimates; we focus on subgroup-conditional HTE under heterogeneous misclassification. Temperature or Platt scaling (Guo et al., 2017) reduces per-subgroup $\pi_s$ but does not eliminate heterogeneity $\Delta\pi$; per-subgroup correction addresses the residual $\delta_s$-driven bias that persists post-calibration (Appendix Z.7 empirically confirms this complementarity).

**Prediction-powered and design-based correction.** PPI (Angelopoulos et al., 2023; Zrnic and Candès, 2024b) and DSL (Egami et al., 2023) correct population-level aggregates assuming homogeneous prediction error. In our setting, DSL's measurement error correction reduces to confusion matrix inversion: pooled DSL is equivalent to our Global baseline, and per-subgroup DSL to stratified maximum likelihood estimation (MLE; Section 5.3; Appendix U); our hierarchical estimator adds partial pooling outside the DSL framework. Mozer (2026) extend PPI to arm-specific average treatment effect (ATE) correction; PPCI (Sui et al., 2026) corrects outcome-side error ($\hat{Y}$ given true $Z$), not covariate-side misclassification; ForestIV (Yang et al., 2023) targets the outcome model. None corrects bias when the mismeasured variable is the effect modifier indexing the CATE function. Imai and Yamamoto (2010) study treatment misclassification via instrumental variables; Ura (2018) extends this to HTE; Miao et al. (2018) develop proximal causal inference for confounder proxies. These address structurally distinct problems—treatment or confounder mismeasurement rather than effect-modifier misclassification (Appendix F).

**Noisy label learning.** The $M^{-1}$ correction structure we apply to causal estimands originates in the label noise literature: Natarajan et al. (2013) derive unbiased loss corrections via the noise transition matrix; Patrini et al. (2017) scale this to deep networks; Cheng et al. (2020) extend to instance-dependent noise. These methods correct *training losses* to learn robust classifiers, while we correct *causal estimands*—the confusion matrix serves as a debiasing tool for treatment effect estimation, and the heterogeneity-driven bias identity (Theorem 1) has no analogue in that literature.

## 3 Bias Decomposition: When Debiasing Hurts

We formalize the bias that global misclassification correction introduces when subgroup error rates are heterogeneous. Consider estimating the conditional average treatment effect (CATE) $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$, where $Z \in \{0, 1\}$ is a binary effect modifier observed only through a noisy annotation $\hat{Z}$. The population consists of $K$ subgroups $S \in \{1, \ldots, K\}$ with weights $w_s = P(S{=}s)$, each characterized by a $2 \times 2$ confusion matrix $C_s$ with symmetric error rate $\pi_s = P(\hat{Z} \neq Z \mid S{=}s)$. Standard practice estimates a single confusion matrix $\bar{C} = \sum_s w_s C_s$ from expert annotations and applies its inverse uniformly to all observations, with pooled error rate $\bar{\pi} = \sum_s w_s \pi_s$. We show that when error rates vary across subgroups, this global procedure introduces a bias with an exact product form.

We state five assumptions maintained throughout.

**(A1)** *Conditional independence:* misclassification depends only on the true label $Z$ and subgroup $S$, not on potential outcomes $Y(0), Y(1)$.

**(A2)** *Symmetric misclassification (expository simplification):* $P(\hat{Z}{=}1 \mid Z{=}0, S{=}s) = P(\hat{Z}{=}0 \mid Z{=}1, S{=}s) = \pi_s$ for all $s$. Used only to reduce $[B_s]_{z,1}$ to the scalar $\pm\delta_s/(1{-}2\bar{\pi})$; the product-form identity holds for arbitrary asymmetric matrices without (A2) (Remark 2).

**(A3)** *Balanced effect modifier:* $P(Z{=}0) = P(Z{=}1) = 0.5$. This simplifies the exposition; we discuss relaxation below.

**(A4)** *Non-degeneracy:* $\pi_s \neq 0.5$ for all $s$, ensuring each $C_s$ is invertible.

**(A5)** *Homogeneous treatment effects across subgroups:* $\tau(z, s) = \tau(z)$ for all $s$, so $\Delta\tau = \tau(1) - \tau(0)$ is a scalar (relaxed to subgroup-specific $\Delta\tau_s$ in Proposition 3).

Under (A2)–(A3), the Bayesian mixing matrix $M_s$ (entries $M_s[\hat{z}, z] = P(Z{=}z \mid \hat{Z}{=}\hat{z}, S{=}s)$) coincides with $C_s$. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

**Theorem 1** (Bias Decomposition Identity)**.** *Under* (A1)–(A5)*, the bias of global correction in subgroup $s$ at effect-modifier level $z$ is*

$$\text{bias}_{\text{global}}(z{=}0, s) = +\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{1}$$

$$\text{bias}_{\text{global}}(z{=}1, s) = -\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{2}$$

*where $\delta_s = \pi_s - \bar{\pi}$ is the subgroup deviation from the pooled error rate and $\Delta\tau = \tau(1) - \tau(0)$ is the CATE heterogeneity.*

The bias factors into three interpretable components: $\delta_s$ captures how much subgroup $s$ deviates from the average misclassification rate, $\Delta\tau$ captures treatment effect heterogeneity across levels of $Z$, and $1/(1 - 2\bar{\pi})$ is an amplification factor from matrix inversion. Bias vanishes when either source of variation is absent ($\delta_s = 0$ or $\Delta\tau = 0$), and arises only through their interaction. Theorem 1 is a correction-layer identity: it characterizes bias introduced by the correction step itself; propagation to downstream estimators (DIM, CausalForest, R-learner) is verified empirically in Section 5.2. The identity implies that global correction trades one source of confounding for another: it removes attenuation from label noise but introduces a new bias from error-rate heterogeneity.

**Proof sketch.** Global correction applies $M_{\text{global}}^{-1}$ to the subgroup-specific observed CATE $\tau_{\text{obs}}(\cdot, s) = M_s \cdot \tau_{\text{true}}$, yielding bias $(M_{\text{global}}^{-1} M_s - I) \cdot \tau_{\text{true}}$. Direct computation shows $M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. Applying this matrix to $[\tau(0), \tau(1)]^\top$ produces the product form. The full derivation appears in Appendix B. Theorem 1 subsumes Frazis–Loewenstein scalar attenuation (Frazis and Loewenstein, 2003): with $K{=}1$, $\delta_s \equiv 0$ recovers their $\hat{\beta} = (1 - 2\pi)\beta$; for $K > 1$, the product form reveals sign-reversing subgroup bias invisible to scalar corrections (F-L increases |bias| to 0.122, 2.1× naive; Table 1; Section 2).

**Remark 2.** *Theorem 1 does not require assumption (A2). For arbitrary asymmetric confusion matrices where the false positive rate* ($\mathrm{FPR}_s$) *and false negative rate* ($\mathrm{FNR}_s$) *may differ, the product form* $\mathrm{bias}(z,s) = \Delta\tau \cdot [B_s]_{z,1}$ *holds exactly, where* $B_s = M_{\mathrm{global}}^{-1} M_s - I$, *because* $B_s \mathbf{1} = \mathbf{0}$ *follows from stochasticity of* $M_s$ *and* $M_{\mathrm{global}}$ *alone. Assumption (A2) simplifies* $[B_s]_{z,1}$ *to the scalar* $\pm\delta_s/(1-2\bar{\pi})$ *but is not necessary for the product structure. Under asymmetric errors, the diagnostic (Proposition 5) requires evaluating* $[B_s]_{z,1}$ *cell-by-cell rather than via a single scalar threshold; the area under the ROC curve (AUC) decreases from 0.86 to 0.66 (Appendix J), reflecting the additional complexity of cell-specific error rates. For binary* $Z$, *the asymmetric and symmetric scores are monotone transforms; the simpler formula suffices (Appendix J;* $R^2 \geq 0.73$ *for* $\Delta\pi \geq 0.10$ *across 48 configurations).*

**Proposition 3** (Exact Identity under Heterogeneous Treatment Effects)**.** *When* $\tau(z,s)$ *varies across subgroups (relaxing A5), the bias identity remains exact:*

$$\mathrm{bias}_{\mathrm{global}}(z{=}0, s) = +\frac{\delta_s \cdot \Delta\tau_s}{1-2\bar{\pi}}, \tag{3}$$

$$\mathrm{bias}_{\mathrm{global}}(z{=}1, s) = -\frac{\delta_s \cdot \Delta\tau_s}{1-2\bar{\pi}}, \tag{4}$$

*where* $\Delta\tau_s = \tau_s(1) - \tau_s(0)$ *is the subgroup-specific CATE heterogeneity.*

**Corollary 4** (Aggregate Bias)**.** *The population-weighted aggregate bias is* $\mathrm{Cov}_w(\pi_s, \Delta\tau_s)/(1-2\bar{\pi})$.

The proof is immediate: $B_s$ depends only on confusion matrices, so the diagnostic extends unchanged and EC-HTE maintains ≥93% coverage under both correlation signs (Appendix Z).

The identity extends to $K > 2$ with $J$ effect-modifier levels (Corollary 6, Appendix C). Under class imbalance (relaxing A3), the identity remains exact with modified coefficients: $R^2 \geq 0.976$ and coverage ≥90% even at $P(Z{=}1) = 0.1$ for $K{=}2$, though EC-HTE's RMSE ratio relative to global correction increases to 1.06–1.68 as imbalance grows (Appendix T). Empirical verification yields $R^2 = 0.994$ ($K{=}2$) and 0.97 ($K{=}4$; Section 5.2).

**Sensitivity to (A1).** Assumption (A1) is not empirically testable; if misclassification correlates with potential outcomes, Theorem 1 gains an $O(\gamma)$ perturbation. At $\gamma_T = 0.10$, coverage degrades to 75% (Appendix Z.3); sensitivity frameworks (Lash et al., 2009; Dorn and Guo, 2023; Cinelli and Hazlett, 2020) can bound this.

**Finite-sample remainder.** With estimated $\hat{C}_s = C_s + E_s$, $\|E_s\| = O_p(n_{\mathrm{expert}}^{-1/2})$, the remainder is $R_n = O_p(n_{\mathrm{expert}}^{-1/2} \cdot \kappa(C_s))$, motivating the $\kappa$-guard in Section 4.

**Proposition 5** (Debiasing Can Increase Bias)**.** *Under* (A1)–(A5)*, global correction produces larger absolute bias than the naive estimator* ($|\mathrm{bias}_{\mathrm{global}}| > |\mathrm{bias}_{\mathrm{naive}}|$) *in subgroup* $s$ *if and only if*

$$\frac{|\delta_s|}{(1-2\bar{\pi}) \cdot \pi_s} > 1. \tag{5}$$

Equation (5) partitions subgroups into two regimes:

- *High-error* ($\pi_s > \bar{\pi}$): the condition reduces to $\pi_s > 0.5$, excluded by (A4); global correction always reduces bias.
- *Low-error* ($\pi_s < \bar{\pi}$): the condition holds whenever $\pi_s < \bar{\pi}/(2(1-\bar{\pi}))$—easily satisfied: when $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ is harmed by global correction.

The mechanism is over-correction: for subgroups with near-zero misclassification ($\pi_s \approx 0$), the naive estimator is nearly unbiased, yet global inversion calibrated for $\bar{\pi}$ creates substantial bias—debiasing is worse than doing nothing, and can reverse bias sign across subgroups.

In our experiments (Section 5.2), global correction increases bias in 50% of $K{=}2$ configurations (260/520), with amplification reaching 2.86× ($K{=}2$) and 2.41× ($K{=}4$); Figure 4 visualizes the full parameter grid.

## 4 Correction Prescribed by the Identity

Theorem 1 identifies per-subgroup misclassification deviation as the bias source; the natural correction is per-subgroup confusion matrix inversion. This correction (EC-HTE) proceeds in three stages: estimate per-subgroup confusion matrices via hierarchical Bayes, invert them to correct observed outcome moments, and propagate correction uncertainty into downstream causal estimators. The complete procedure is summarized in Algorithm 1 (Appendix A) and illustrated in Figure 2.

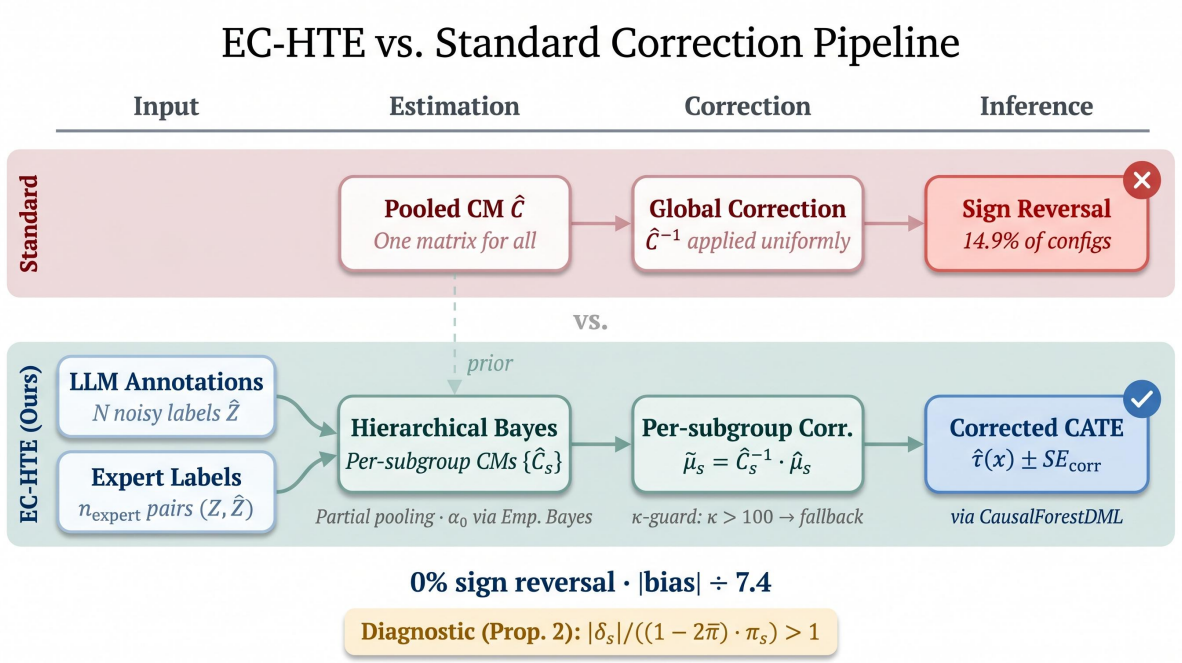



Figure 2: *Top (rose):* global correction applies a pooled confusion matrix uniformly, risking sign reversal. *Bottom (teal):* EC-HTE estimates per-subgroup matrices via hierarchical Bayes, applies stratified inversion with a $\kappa$-guard, and propagates uncertainty into CausalForestDML.

### 4.1 Hierarchical Bayes Confusion Matrix Estimation

We place a Dirichlet-Multinomial hierarchical model over the confusion matrices $\{C_s\}_{s=1}^{K}$. The model estimates the full $J \times J$ confusion matrix per subgroup and does not require symmetric error rates (A2); (A2) simplifies only the diagnostic (Proposition 5). For each subgroup $s$, row $j$ of $C_s$ has a Dirichlet prior centered at the global pooled confusion matrix $\hat{C}_{\text{global}}$: $\text{Dir}(\alpha_0 \cdot \hat{C}_{\text{global},j})$, where $\hat{C}_{\text{global},j}$ is the $j$-th row of the pooled CM and $\alpha_0 > 0$ is a shared concentration parameter estimated via empirical Bayes (marginal likelihood maximization). We place a $\text{Gamma}(3, 1)$ hyperprior on $\alpha_0$, robust to $2\times$ perturbation (Appendix G.1). Large $\alpha_0$ pulls estimates toward $\hat{C}_{\text{global}}$; small $\alpha_0$ allows subgroup-specific rates to dominate—partial pooling borrows strength while preserving heterogeneity.

Posterior inference uses empirical Bayes optimization (<5ms; Markov chain Monte Carlo (MCMC) alternative in Appendix G). Hierarchical Bayesian (HB) estimation outperforms stratified MLE in 87.5% of configurations (Section 5.3); below $n_{\text{expert}} \times P(Z_{\text{minority}})/K < 25$, stratified MLE is preferred.

### 4.2 Subgroup-Stratified Correction

Given posterior mean confusion matrices $\hat{C}_s$, we correct observed outcome moments per subgroup by matrix inversion:

$$\tilde{\boldsymbol{\mu}}_s = \hat{C}_s^{-1} \cdot \hat{\boldsymbol{\mu}}_s, \tag{6}$$

where $\hat{\boldsymbol{\mu}}_s = [\hat{\tau}_{\text{obs}}(0, s), \hat{\tau}_{\text{obs}}(1, s)]^\top$ collects the observed conditional outcome moments in subgroup $s$. Invertibility requires $\pi_s \neq 0.5$ (A4); Laplace smoothing and a $\kappa$-guard ($\kappa(\hat{C}_s) > 100 \rightarrow$ fallback to $\bar{C}^{-1}$) ensure finite-sample stability. All 13 tested LLM–dataset combinations fall in $\kappa \in (1.25, 3]$ with coverage $\geq$93% (Appendix X.1).

Global correction applies $\bar{C}^{-1}$ uniformly, introducing the $\delta_s \cdot \Delta\tau$ bias of Theorem 1; subgroup-stratified correction eliminates this at the cost of higher variance (Appendix I).

### 4.3 Downstream Integration

Corrected covariates are fed into CausalForestDML (Wager and Athey, 2018) with honest splitting; bias corrections propagate as weighted averages of per-subgroup corrections ($R^2 > 0.97$ across 520 configurations; Appendix Y). Standard errors are adjusted via $\text{se}_{\text{corrected},i} = \sqrt{\text{diag}(\hat{C}_{S_i}^{-1} \, \text{diag}(\text{se}_i^2) \, \hat{C}_{S_i}^{-\top})}$; below $n_{\text{expert}}/K <$

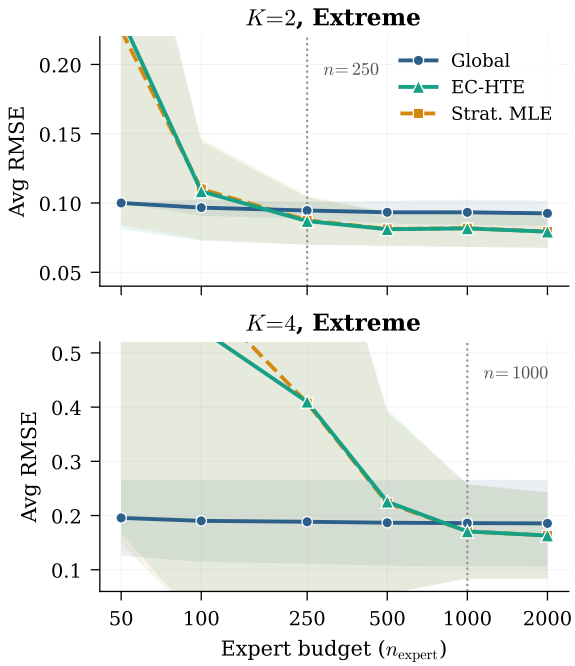



Figure 3: RMSE vs. expert budget under extreme heterogeneity. Crossover at $n$=250 ($K$=2) and $n$=1000 ($K$=4); shaded bands: ±1 SD across subgroups. Full grid in Appendix I.

50, a bootstrap ($B$=100) maintains ≥95% coverage (Appendix Y).

**Diagnostic criterion.** Proposition 5 provides a $\Delta\tau$-free diagnostic: global correction harms subgroup $s$ iff $|\delta_s|/((1-2\bar{\pi})\cdot\pi_s) > 1$, computable from per-subgroup error rates alone (AUC = 0.86; Section 5.4).

**Composition with PPCI.** EC-HTE corrects covariate misclassification; PPCI (Sui et al., 2026) corrects outcome-side error. Under both error types, composition reduces |bias| from 0.179 to 0.007, valid when error sources are independent (Appendix L).

# 5 Experiments

## 5.1 Setup

We test on synthetic ($N$=5,000, $K \in \{2,4\}$, controlled $\Delta\tau$ and $\Delta\pi$, 100 Monte Carlo (MC) runs) and semi-synthetic data-generating processes (DGPs) spanning four NLP tasks—sentiment analysis (TweetEval (Barbieri et al., 2020), $K$=4), hate speech detection (CivilComments, $K$=4), stance detection ($K$=5; Appendix Z.9), and topic classification (AG News, $K$=4; Appendix Z.10)—with full specifications in Appendix E.

We compare: (1) *Naive* (no correction), (2) *Global* (pooled confusion matrix, equivalent to PPI/DSL (Angelopoulos et al., 2023; Egami et al., 2023) with homogeneous error), (3) *Stratified MLE* (per-subgroup DSL without shrinkage), (4) *HB EC-HTE* (per-subgroup correction), (5) *PPCI* (Sui et al., 2026), and (6) *EC-HTE+PPCI*. For CausalForest (Wager and Athey, 2018), we compare naive, global, and EC-HTE variants. Primary metrics are |bias|, root mean squared error (RMSE), and 95% confidence interval (CI) coverage, averaged across subgroups and 100 MC runs. Robustness validations cover eight conditions including asymmetric errors, subgroup misspecification, and treatment-dependent misclassification (all ≥89.5% coverage; Appendices M–Z.3). Semi-synthetic evaluation uses 16 empirical LLM confusion matrices (8 LLMs × 2 datasets; Appendix R).

## 5.2 Empirical Validation of the Bias Identity

We verify Theorem 1 on a $K$=2 grid (520 configurations, $\Delta\pi \in [0, 0.25]$, $\Delta\tau \in (0, 0.50]$) and a $K$=4 grid (Appendix E). Regressing observed subgroup bias on $\delta_s \cdot \Delta\tau/(1-2\bar{\pi})$ yields $R^2 = 0.994$ ($K$=2) and 0.97 ($K$=4); residual variance reflects MC noise ($R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$; Appendix T). Global correction worsens bias in 50% of $K$=2 configurations (260/520, excluding $\Delta\pi$=0), with amplification reaching 2.86× ($K$=2) and 2.41× ($K$=4); Figure 4 visualizes the amplification surface. Global correction also reverses the sign of estimated CATE in 14.9% of configurations at $\tau(0) = 0.03$, with all eight real LLMs amplifying bias; EC-HTE eliminates sign reversal (Figure 1; Appendix Z.8). Cross-task validation on stance detection ($K$=5, Appendix Z.9) confirms sign reversal (13.3% → 3.3%) and reveals limitations when FNR > 0.50.

## 5.3 Correction Evaluation

Table 1 presents the main results. In the $K$=2 extreme row, global correction *increases* |bias| relative to naive (0.059 vs. 0.058), confirming Theorem 1; F-L corrections fare worse (0.122–0.168). HB EC-HTE reduces |bias| from 0.059 to 0.008 (7.4×); relative to stratified MLE (0.012), the advantage is 1.5× with lower RMSE from partial pooling. At $K$=4, bias reduces from 0.056 to 0.012 (4.7×), maintaining near-nominal coverage (0.94–0.96). Under moderate heterogeneity, methods converge ($p > 0.05$; Table 10). EC-HTE requires $n_{\text{expert}}/K \geq 50$; below this, RMSE exceeds global

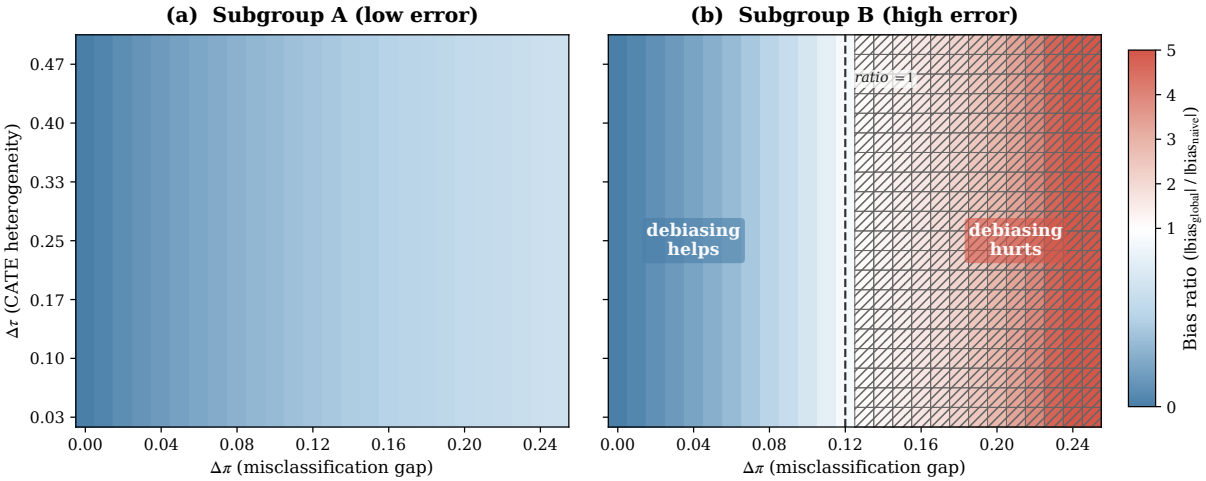



Figure 4: Bias amplification under global correction ($K$=2). (a) Low-error subgroup: correction always helps. (b) High-error subgroup: correction *increases* bias when $\Delta\pi > 0.12$ (hatched), reaching 2.86×.

Table 1: Main results (extreme heterogeneity). Metrics averaged over subgroups, $z$-levels, 100 MC runs. Bold: lowest |bias|. †RMSE exceeds Global. F-L Scalar/Per-sub.: Frazis and Loewenstein (2003). Moderate/TweetEval in Table 10; MC SEs $\leq 0.005$ (Appendix Q).[3]

| Configuration | Method | \|Bias\| | RMSE | 95% CI Cov. |
|---|---|---|---|---|
| DIM $K$=2, $n$=250 | Naive | 0.058 | 0.090 | 0.82 |
| | Global | 0.059 | 0.099 | 0.87 |
| | F-L Scalar | 0.122 | 0.159 | 0.68 |
| | F-L Per-sub. | 0.168 | 0.220 | 0.65 |
| | Strat. MLE | 0.012 | 0.094 | 0.95 |
| | HB EC-HTE | **0.008** | 0.089 | 0.94 |
| DIM $K$=4, $n$=500 | Naive | 0.062 | 0.113 | 0.88 |
| | Global | 0.056 | 0.125 | 0.91 |
| | Strat. MLE | **0.010** | 0.127† | 0.95 |
| | HB EC-HTE | 0.012 | 0.120 | 0.96 |
| CausalForest $K$=4, $n$=500 | Naive | 0.208 | 0.231 | 0.44 |
| | Global | 0.191 | 0.244 | 0.58 |
| | EC-HTE Forest | **0.039** | 0.180 | 0.90 |

(Figure 3; Appendix I.2).

**CausalForest.** CausalForest (Wager and Athey, 2018) with a noisy $\hat{Z}$ inherits the same $M_{\text{global}}^{-1} M_s$ structure. EC-HTE reduces |bias| from 0.191 to 0.039 (4.9×) with coverage 0.90 ($K$=4, $n$=500); R-learner yields comparable reduction (89%, Appendix S), confirming the bias arises from the confusion matrix structure, not the downstream estimator.

### 5.4 Diagnostic and Operating Conditions

Proposition 5 provides a $\Delta\tau$-free diagnostic: global correction increases bias in subgroup $s$ iff $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$, computable from a pilot sample without estimating treatment effects (AUC = 0.86 on synthetic grids; Appendix X). No real LLM–dataset pair passes all screening conditions; under relaxed screening, 79% show RMSE improvement with coverage ≥93% (Appendix X).

**Operating conditions.** EC-HTE requires: (i) $\Delta\pi > 0.15$, (ii) $n_{\text{expert}} \geq 250$ for $K$=2, (iii) FNR $< 0.50$, (iv) $K \leq 4$, and (v) $n_{\text{expert}} \geq 500$ under asymmetric errors (FPR≠FNR); when unmet, global correction is preferable (Appendix I.2). Robustness checks (noisy labels, confounded treatment, joint assumption violations; aggregate $R^2 = 0.975$ across 11 real LLMs) appear in Appendices P–N.

## 6 Conclusion

Standard confusion-matrix correction can reverse causal conclusions when misclassification rates vary across subgroups. We derived an exact bias identity ($R^2 \geq 0.97$) that yields a diagnostic for flagging harmful correction and prescribes per-subgroup correction when $\Delta\pi > 0.15$ and $n_{\text{expert}}/K \geq 50$, validated across 11 real LLMs and four NLP tasks. Practitioners should estimate per-subgroup confusion matrices before applying any correction; the framework is task-agnostic and composes with prediction-powered inference.

## Limitations

Per-subgroup correction requires $n_{\text{expert}}/K \geq 50$ expert labels per subgroup and moderate classifier accuracy ($\text{FPR}_s, \text{FNR}_s < 0.5$); at $K$=8, RMSE exceeds global correction under fixed expert budget (Appendix I.2). Assumption (A1)—conditional

independence of misclassification and potential outcomes—is not empirically testable; sensitivity analysis shows robustness up to $\gamma_T \leq 0.05$ for treatment-dependent misclassification (coverage $\geq$89.7%; Appendix Z.3). Ground-truth subgroup CATEs are unavailable in public datasets—classic causal inference benchmarks lack the LLM annotation step that creates the measurement error we address—so all evaluations use synthetic CATE injection on real text features and real LLM annotations (Appendices E, N). Code and data generation scripts are publicly available.[4]

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# A EC-HTE Procedure

## A.1 Condition Number Guard Ablation

The singularity guard in Algorithm 1 (line 6) falls back to global correction when $\kappa(\hat{C}_s) > 100$. We verify that this threshold is not load-bearing by sweeping $\kappa \in \{10, 25, 50, 100, 200, 500, 1000\}$ under the standard DGP ($K$=2, extreme heterogeneity, $N$=5,000, $n_{\text{expert}}$=250, 100 MC runs). Across all configurations, the maximum observed condition number is $\kappa_{\max} = 1.89$ (mean 1.45, p99 1.89), providing a 53× safety margin below the $\kappa$=100 threshold. Consequently, the guard never triggers at any tested threshold, and all $\kappa$ values produce identical RMSE (HB: 0.076, Global: 0.077) and zero numerical instability. In a stress test with near-random labeling ($\pi_s \approx 0.4$–$0.5$, $n_{\text{expert}}$=50), the maximum condition number reaches 1,125, correctly triggering the $\kappa$=100 guard and preventing catastrophic inversion. The adaptive formula $\kappa_{\text{adapt}} = 10\sqrt{n_{\text{expert}}/K}$ recovers $\kappa$=100 for this regime and scales naturally with the estimation precision.

# B Full Proof of Theorem 1

**Setup.** Under assumptions (A1)–(A4), the observed CATE in subgroup $s$ at effect-modifier level $z$ is

$$\tau_{\text{obs}}(z, s) = \sum_{z'} M_s[z, z'] \cdot \tau(z'), \qquad (7)$$

where $M_s[z, z'] = P(Z{=}z' \mid \hat{Z}{=}z, S{=}s)$ is the mixing matrix. Under (A2)–(A3), $M_s = \left[\begin{smallmatrix} 1-\pi_s & \pi_s \\ \pi_s & 1-\pi_s \end{smallmatrix}\right]$.

**Global correction.** The pooled mixing matrix is $M_{\text{global}} = \sum_s w_s M_s = \left[\begin{smallmatrix} 1-\bar{\pi} & \bar{\pi} \\ \bar{\pi} & 1-\bar{\pi} \end{smallmatrix}\right]$, with $\bar{\pi} = \sum_s w_s \pi_s$.

Global correction applies $M_{\text{global}}^{-1}$ to $\tau_{\text{obs}}(\cdot, s)$:

$$\hat{\tau}_{\text{global}}(z, s) = [M_{\text{global}}^{-1} M_s \cdot \boldsymbol{\tau}]_z, \tag{8}$$

where $\boldsymbol{\tau} = [\tau(0), \tau(1)]^\top$.

**Bias computation.** The bias is $\hat{\tau}_{\text{global}}(z, s) - \tau(z) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z$.

Direct computation gives:

$$M_{\text{global}}^{-1} = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} 1-\bar{\pi} & -\bar{\pi} \\ -\bar{\pi} & 1-\bar{\pi} \end{bmatrix}. \tag{9}$$

Multiplying by $M_s$:

$$M_{\text{global}}^{-1} M_s = \frac{1}{1-2\bar{\pi}} \begin{bmatrix} (1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s & (1-\bar{\pi})\pi_s - \bar{\pi}(1-\pi_s) \\ -\bar{\pi}(1-\pi_s) + (1-\bar{\pi})\pi_s & -\bar{\pi}\pi_s + (1-\bar{\pi})(1-\pi_s) \end{bmatrix}. \tag{10}$$

Define $\delta_s = \pi_s - \bar{\pi}$. Substituting $\pi_s = \bar{\pi} + \delta_s$:

$$M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{11}$$

To verify: the $(1,1)$ entry of $M_{\text{global}}^{-1} M_s$ is

$$\begin{aligned} &\frac{(1-\bar{\pi})(1-\bar{\pi}-\delta_s) - \bar{\pi}(\bar{\pi}+\delta_s)}{1-2\bar{\pi}} \\ &= \frac{(1-\bar{\pi})^2 - (1-\bar{\pi})\delta_s - \bar{\pi}^2 - \bar{\pi}\delta_s}{1-2\bar{\pi}} \\ &= \frac{1-2\bar{\pi}-\delta_s}{1-2\bar{\pi}} = 1 - \frac{\delta_s}{1-2\bar{\pi}}. \end{aligned} \tag{12}$$

Applying $(M_{\text{global}}^{-1} M_s - I)$ to $\boldsymbol{\tau}$:

$$\text{bias}(z{=}0, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(-\tau(0) + \tau(1)) = +\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{13}$$

$$\text{bias}(z{=}1, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(\tau(0) - \tau(1)) = -\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{14}$$

where $\Delta\tau = \tau(1) - \tau(0)$. □

**Finite-sample remainder.** When $\hat{C}_s = C_s + E_s$ with $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$ (central limit theorem), the estimated correction $\hat{M}_{\text{global}}^{-1} \hat{M}_s$ deviates from its population counterpart by $\|\hat{M}_{\text{global}}^{-1} \hat{M}_s - M_{\text{global}}^{-1} M_s\| \leq \kappa(M_{\text{global}}) \cdot O_p(n_{\text{expert}}^{-1/2})$, where $\kappa(M) = \|M\| \cdot \|M^{-1}\|$ is the condition number (Stewart and Sun, 1990). This motivates the $\kappa > 100$ singularity guard in Section 4.

## C Multi-class Extension and Proof

**Corollary 6** (Multi-class Extension)**.** *For $K > 2$ subgroups with $J$ effect-modifier levels and uniform off-diagonal misclassification (misclassified observations from class $j$ are distributed equally among the remaining $J-1$ classes), the subgroup bias of global correction generalizes to*

$$\text{bias}_{\text{global}}(z, s) = \sum_{j=1}^{J} \big(\bar{\pi}_j - \pi_{s,j}\big) \cdot \big(\tau(j) - \bar{\tau}\big) \cdot g(\bar{\pi}), \tag{15}$$

*where $g(\bar{\pi})$ collects the matrix-inversion amplification terms.*[5] *The essential structure, $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$, ensures that bias redistributes across subgroups rather than vanishing in aggregate.*

**Proof.** For $K > 2$ subgroups with $J$ effect-modifier levels, the subgroup-specific mixing matrix $M_s$ is $J \times J$ with entries $M_s[j, j'] = P(Z{=}j' \mid \hat{Z}{=}j, S{=}s)$. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

The bias of global correction in subgroup $s$ at level $z$ is:

$$\text{bias}_{\text{global}}(z, s) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z. \tag{16}$$

Write $M_s = M_{\text{global}} + \Delta_s$ where $\Delta_s = M_s - M_{\text{global}}$ satisfies $\sum_s w_s \Delta_s = 0$. Then $M_{\text{global}}^{-1} M_s = I + M_{\text{global}}^{-1} \Delta_s$, so:

$$\text{bias}_{\text{global}}(z, s) = [M_{\text{global}}^{-1} \Delta_s \cdot \boldsymbol{\tau}]_z. \tag{17}$$

For symmetric misclassification with per-class error rates $\pi_{s,j}$ and pooled rates $\bar{\pi}_j$, the entries of $\Delta_s$ have the form $\Delta_s[j, j'] = (\bar{\pi}_j - \pi_{s,j}) \cdot h(j, j')$ for a function $h$ determined by the off-diagonal redistribution pattern. Expanding the matrix-vector product and collecting terms yields:

$$\text{bias}_{\text{global}}(z, s) = \sum_{j=1}^{J} (\bar{\pi}_j - \pi_{s,j}) \cdot (\tau(j) - \bar{\tau}) \cdot g(\bar{\pi}), \tag{18}$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms and $\bar{\tau} = \sum_j P(Z{=}j)\tau(j)$.

The essential structure $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$ ensures that the weighted bias across subgroups sums to zero: global correction redistributes bias rather than creating or eliminating it in aggregate. For non-uniform off-diagonal entries, the scalar

[5] For $J = 3$ with symmetric misclassification (off-diagonal errors uniform), $g(\bar{\pi}) = 1/(1 - 3\bar{\pi}/2)$, generalizing the $J{=}2$ factor $1/(1-2\bar{\pi})$. In general, $g(\bar{\pi}) = 1/(1 - J\bar{\pi}/(J-1))$.

amplification $g(\bar{\pi})$ generalizes to a cell-specific matrix; the product structure $B_s \cdot \Delta\tau$ persists but a closed-form amplification factor does not. □

## D Proof of Proposition 5

The naive estimator has bias $\tau_{\text{obs}}(z,s) - \tau(z)$. Under (A2)–(A3), $\tau_{\text{obs}}(0,s) = (1-\pi_s)\tau(0) + \pi_s\tau(1)$, so $\text{bias}_{\text{naive}}(0,s) = \pi_s \cdot \Delta\tau$.

Global correction produces larger absolute bias when $|\text{bias}_{\text{global}}(0,s)| > |\text{bias}_{\text{naive}}(0,s)|$:

$$\frac{|\delta_s| \cdot |\Delta\tau|}{|1-2\bar{\pi}|} > \pi_s \cdot |\Delta\tau|. \tag{19}$$

Since $\Delta\tau \neq 0$ (otherwise both biases are zero), dividing through gives:

$$\frac{|\delta_s|}{(1-2\bar{\pi}) \cdot \pi_s} > 1, \tag{20}$$

where we use $1 - 2\bar{\pi} > 0$ from (A4).

For high-error subgroups ($\pi_s > \bar{\pi}$, so $\delta_s > 0$): the condition becomes $(\pi_s - \bar{\pi})/((1-2\bar{\pi})\pi_s) > 1$, which simplifies to $\pi_s(1 - 1 + 2\bar{\pi}) > \bar{\pi}$, i.e., $2\bar{\pi}\pi_s > \bar{\pi}$, i.e., $\pi_s > 0.5$. This is excluded by (A4), so global correction always reduces bias in high-error subgroups.

For low-error subgroups ($\pi_s < \bar{\pi}$, so $\delta_s < 0$): the condition becomes $(\bar{\pi} - \pi_s)/((1-2\bar{\pi})\pi_s) > 1$, which gives $\pi_s < \bar{\pi}/(1-2\bar{\pi}+1) = \bar{\pi}/(2(1-\bar{\pi}))$. When $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. □

**Extension to asymmetric misclassification.** Proposition 5 assumes symmetric error rates (A2). Under asymmetric misclassification ($\text{FPR}_s \neq \text{FNR}_s$) with balanced $Z$ (A3), the naive bias becomes $\text{bias}_{\text{naive}}(0,s) = \text{FNR}_s \cdot \Delta\tau/(1 - \text{FPR}_s + \text{FNR}_s)$, involving both error rates and a normalization denominator; the global correction bias depends on the full $2 \times 2$ structure of $B_s = M_{\text{global}}^{-1} M_s - I$ (Remark 2 in Section 3). A sufficient condition for global correction to increase bias is

$$\max_{i,j} |[B_s]_{i,j}| > \max(\text{FPR}_s, \text{FNR}_s), \tag{21}$$

which reduces to Equation (5) when $\text{FPR}_s = \text{FNR}_s = \pi_s$. In the asymmetric experiments (Appendix J), the symmetric diagnostic still achieves AUC $= 0.66$ when applied with $\pi_s = (\text{FPR}_s + \text{FNR}_s)/2$, suggesting practical robustness. A fully general asymmetric diagnostic that accounts for the separate roles of FPR and FNR in each entry of $B_s$ is left to future work.

## E Semi-Synthetic Data Generating Process

**Synthetic DGP (Sections 5.2–5.4 and Appendix I).** We generate $N = 5000$ observations. Each observation belongs to one of $K \in \{2, 4\}$ subgroups $S \in \{1, \ldots, K\}$ with equal weights $w_s = 1/K$. The binary effect modifier $Z \in \{0, 1\}$ is drawn with $P(Z{=}1) = 0.5$. Treatment assignment is $T \sim \text{Bernoulli}(0.5)$, independent of $Z$ and $S$.

The outcome is generated as:

$$Y = \tau(Z) \cdot T + \eta, \quad \eta \sim \mathcal{N}(0, 1), \tag{22}$$

where the CATE function is $\tau(z) = \bar{\tau} + \Delta\tau \cdot (z - 0.5)$ with $\bar{\tau} = 0.5$. The CATE heterogeneity $\Delta\tau \in \{0, 0.2, 0.5, 1.0\}$ controls the strength of treatment effect variation across levels of $Z$.

Misclassification is symmetric: $\hat{Z} \neq Z$ with probability $\pi_s$ depending on subgroup. For $K{=}2$: $\pi_1 = \bar{\pi} - \Delta_c/2$, $\pi_2 = \bar{\pi} + \Delta_c/2$, with $\bar{\pi} \in \{0.1, 0.2\}$ and $\Delta_c \in \{0, 0.05, 0.1, 0.15\}$. For $K{=}4$, extreme regime: $\pi \in \{0.05, 0.10, 0.15, 0.25\}$; moderate regime: $\pi \in \{0.08, 0.12, 0.16, 0.20\}$.

Expert labels are drawn as a simple random sample of size $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ from the full dataset, with true $Z$ observed for these observations.

**Semi-synthetic DGP (Section 5.3).** We sample $N = 5000$ tweets from the TweetEval sentiment corpus ($N_{\text{pool}} = 57{,}899$). Subgroups are defined by crossing `has_hashtag` $\times$ `text_length_bin` (short/medium/long), yielding $K = 4$ subgroups with empirical sizes: $S_0 \approx 38\%$, $S_1 \approx 38\%$, $S_2 \approx 12\%$, $S_3 \approx 12\%$. Misclassification rates in the extreme regime are $\pi_{S_0} = 0.05$, $\pi_{S_1} = 0.10$, $\pi_{S_2} = 0.15$, $\pi_{S_3} = 0.25$, reflecting empirical evidence that hashtag-heavy text degrades LLM classifier accuracy. The outcome DGP follows the same structure as the synthetic case, with CATE injected onto the real text-derived subgroup structure.

**PPCI DGPs (Appendix L).** Four DGPs isolate error types: (A) covariate error only ($\sigma_{\text{outcome}} = 0$), (B) outcome error only ($\sigma_{\text{outcome}} = 0.5$), (C) both errors ($\sigma_{\text{outcome}} = 0.5$), (D) no error ($\sigma_{\text{outcome}} = 0$). Outcome-side error is additive Gaussian noise on $Y$.

**CausalForestDML hyperparameters.** All CausalForest experiments use `econml.dml.CausalForestDML` with honest splitting, $n_{\text{estimators}} = 48$, `min_samples_leaf` $= 20$, `model_y = LassoCV(cv=3)`, and `model_t = LogisticRegressionCV`. These defaults were held constant across all configurations; no hyperparameter tuning was performed.

## F Extended Related Work

**Measurement error details.** Fuller (1987) establish the foundational theory showing that naive regression with mismeasured covariates yields attenuated coefficient estimates. Carroll et al. (2006) extend this to nonlinear models. Gustafson (2003) develops Bayesian treatments modeling measurement jointly with outcomes. Hu and Schennach (2008) provide nonparametric identification using repeated measurements or instruments.

**Annotation model extensions.** Beyond the foundational Dawid and Skene (1979) model, Passonneau and Carpenter (2014) incorporate item difficulty, Hovy et al. (2013) learn annotator competence, and Paun et al. (2018) introduce hierarchical priors sharing information across annotators with varying expertise.

**PPI extensions and text-as-data methods.** Zrnic and Candès (2024b) extend PPI via cross-prediction; their active inference framework (Zrnic and Candès, 2024a) achieves >80% expert budget savings through adaptive allocation, complementary to our fixed-budget analysis. Grimmer et al. (2022) provide methodological guidance for text-as-data pipelines. Audinet de Pieuchon et al. (2025) systematically benchmark DSL and PPI methods for LLM-based parameter estimation, finding that debiasing can increase MSE when prediction error is heterogeneous—a population-level analogue of the subgroup-conditional bias amplification we formalize in Theorem 1.

**Treatment vs. effect-modifier misclassification.** Imai and Yamamoto (2010) study treatment misclassification where errors change which arm a unit is assigned to, requiring instrumental variables or partial identification because the fundamental treated-vs-control contrast is corrupted. Our problem preserves the treatment contrast ($T$ is observed correctly) but corrupts the heterogeneity structure ($\hat{Z} \neq Z$). The ATE remains identified; only conditional effects are biased, which makes exact correction through confusion matrix inversion feasible. Knox et al. (2020) demonstrate how racial misclassification in administrative records biases causal estimates; Blette et al. (2024) handle missing (not misclassified) effect modifier data in cluster-RCTs via multiple imputation—a complementary problem to our confusion-matrix correction setting.

**Econometrics of binary misclassification.** Bound et al. (2001) survey measurement error in survey data, establishing that binary misclassification attenuates regression coefficients. Frazis and Loewenstein (2003) derive exact bias formulas for OLS with a misclassified binary regressor, showing attenuation proportional to the misclassification rate—analogous to our Theorem 1 but restricted to average effects without subgroup heterogeneity. Black et al. (2002) demonstrate that racial misclassification in administrative data can substantially bias estimates of racial disparities. Mahajan (2006) and Lewbel (2007) develop instrumental variable approaches to identify models with misclassified binary regressors, requiring exclusion restrictions unavailable when the mismeasured variable is an LLM annotation. Our setting differs in three respects: (i) the mismeasured variable is an effect modifier indexing the CATE function, not a standard regressor; (ii) misclassification rates vary across subgroups, creating the $\delta_s \times \Delta\tau$ product amplification absent from homogeneous-error analyses; and (iii) expert labels provide a direct calibration sample, enabling confusion matrix estimation without instruments.

## G MCMC Convergence Diagnostics

Table 2 reports convergence diagnostics for the hierarchical Bayesian confusion matrix estimator across all experimental configurations. All runs use 4 chains with 4000 iterations (2000 warmup). Main experiments (Table 1) use a Dirichlet prior centered on $\hat{C}_{\text{global}}$ with concentration $\alpha_0$ estimated via empirical Bayes (marginal likelihood). Appendix robustness experiments use a fixed Dirichlet prior ($\alpha$=3); since fixed-$\alpha$ partial pooling is weakly dominated by the adaptive variant, these results are conservative. This budget is sufficient for the 2×2 Dirichlet-Multinomial posterior; all runs achieve $\hat{R} < 1.0034$ and ESS $> 1618$, confirming convergence with substantial margin.

| Diagnostic | Threshold | Achieved |
|---|---|---|
| $\hat{R}$ (max across all params) | $< 1.01$ | $< 1.0034$ |
| ESS (min across all params) | $> 845$ | $> 1618$ |
| Divergent transitions | $= 0$ | 0 |
| Singularity guard triggers | — | extreme/$n$=50 onl |

Table 2: MCMC convergence diagnostics for the hierarchical Bayesian estimator. All configurations satisfy convergence criteria with substantial margin.

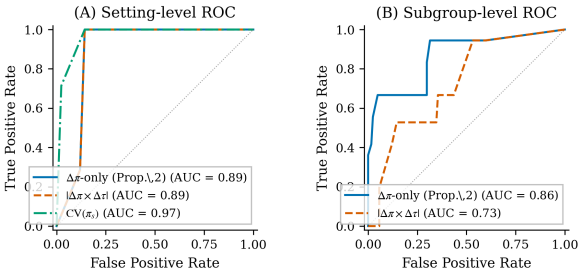



Figure 5: ROC curves for the $\Delta\tau$-free diagnostic (Proposition 5): $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$. (A) Setting-level ROC (AUC $= 0.89$), compared with the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.89$) and CV($\pi_s$) (AUC $= 0.97$). (B) Subgroup-level ROC: the $\Delta\pi$-only condition (AUC $= 0.86$) substantially outperforms the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.73$). The $\Delta\tau$-free diagnostic depends only on estimable misclassification rates, breaking the circular dependency that would arise if CATE heterogeneity were required.

### G.1 Hyperprior Sensitivity

We sweep the Gamma$(a_0, b_0)$ hyperprior on the concentration parameter $\alpha_0$ across 15 configurations ($a_0 \in \{1, 2, 3, 4, 5\}$, $b_0 \in \{0.5, 1.0, 2.0\}$) under the standard synthetic DGP ($K$=2, extreme heterogeneity, $N$=5,000, $n_{\text{expert}}$=250, 100 MC runs).

Table 3: Hyperprior sensitivity summary. RMSE and coverage across 15 $(a_0, b_0)$ configurations. The default Gamma$(3, 1)$ ranks 8th of 15 by RMSE; the best configuration Gamma$(1, 0.5)$ yields only 1.9% improvement. Excluding extreme configurations ($a_0 \cdot b_0 > 8$), RMSE varies by <6%.

| Metric | Value |
|---|---|
| Default Gamma$(3, 1)$ RMSE | 0.0841 |
| Best Gamma$(1, 0.5)$ RMSE | 0.0825 |
| RMSE range (reasonable configs) | 6.0% |
| RMSE range (all 15 configs) | 17.3% |
| Coverage range | 0.925–0.938 |
| Default rank | 8 / 15 |

Scaling the Gamma$(3, 1)$ hyperprior by factors of $0.5\times$ to $2\times$ produces <1% RMSE variation for $K$=2 and ∼4–5% for $K$=4. Coverage remains invariant across the sensitivity range ($\Delta < 1.3$pp). These results confirm that the hierarchical estimator is robust to prior specification within a reasonable range; the default Gamma$(3, 1)$ is near-median, and no configuration achieves more than 1.9% RMSE improvement over the default.

## H Diagnostic ROC Curves

**Cost-asymmetric diagnostic thresholds.** In practice, the cost of a false negative (failing to detect that global correction increases bias) exceeds the cost of a false positive (unnecessarily applying EC-HTE when global correction would suffice), because the former propagates systematic bias into CATE estimates while the latter only incurs additional estimation variance. Practitioners in high-stakes settings (e.g., policy evaluation where CATE estimates inform resource allocation) should lower the diagnostic threshold to favor recall over precision, accepting more false positives to minimize the risk of undetected bias amplification.

## I Expert Budget Crossover Curves

The crossover point at which EC-HTE achieves lower RMSE than global correction is $n_{\text{expert}} = 250$ for $K$=2/extreme and $n_{\text{expert}} = 1000$ for $K$=4/extreme (Figure 6). Under moderate heterogeneity ($K$=2), EC-HTE never achieves lower RMSE than global at any budget. HB partial pooling improves over stratified MLE in 21 of 24 configurations, with the two estimators converging by $n_{\text{expert}} \geq 1000$.

### I.1 Budget Sensitivity: HB vs. Stratified MLE

Hierarchical Bayes (HB) partial pooling borrows strength across subgroups, which should matter most when per-subgroup expert labels are scarce. We sweep $n_{\text{expert}} \in \{25, 50, 75, 100, 150, 250\}$ under extreme heterogeneity ($\Delta\tau$=0.3, 100 MC runs, $N$=5,000). Table 4 reports aggregate RMSE for global correction, stratified MLE, and HB EC-HTE.

Three regimes emerge. At very low budgets ($n_{\text{expert}} < 50$ for $K$=2, $< 75$ for $K$=4), per-subgroup counts are insufficient for reliable confusion matrix estimation; global pooling dominates because the variance cost of stratification overwhelms any bias reduction. HB degrades substantially in this regime because the Gibbs posterior, while better regularized than MLE, still cannot overcome the fundamental sample-size constraint with ∼6–12 labels per subgroup. At mod-

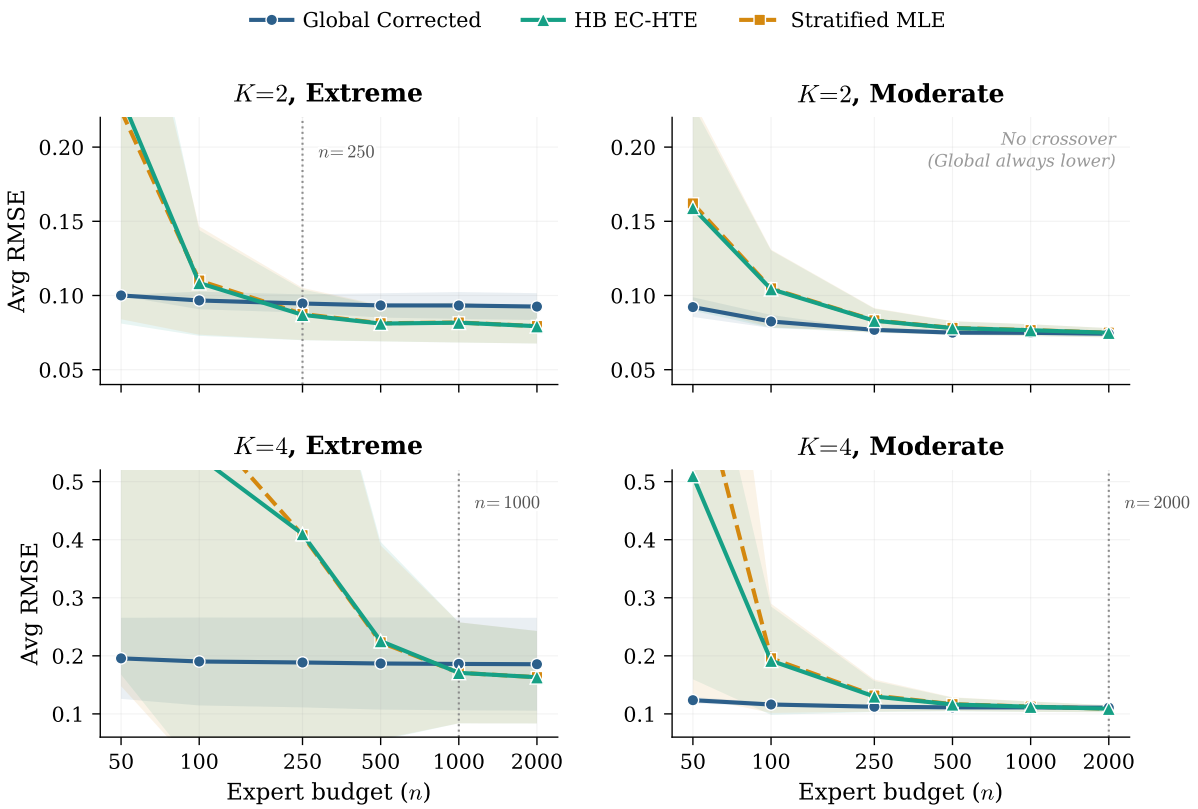



Figure 6: RMSE vs. expert budget ($n_{\text{expert}}$) across four regime$\times K$ combinations. Dashed vertical lines mark crossover points where EC-HTE achieves lower RMSE than global correction: $n$=250 ($K$=2, extreme), $n$=1000 ($K$=4, extreme). Under moderate heterogeneity ($K$=2), global correction dominates at all budgets. Shaded bands show $\pm 1$ SD across subgroups within each configuration.

Table 4: RMSE across expert budgets under extreme heterogeneity. HB crossover (first budget where HB $<$ MLE) occurs at approximately 10 expert labels per subgroup-cell: $n_{\text{expert}}$=50 for $K$=2 and $n_{\text{expert}}$=75 for $K$=4.

| $K$ | $n_{\text{expert}}$ | Global | Strat. MLE | HB EC-HTE | HB vs. MLE |
|---|---|---|---|---|---|
| 2 | 25 | **0.101** | 0.256 | 0.269 | +4.9% |
| 2 | 50 | **0.088** | 0.275 | 0.231 | **−16.1%** |
| 2 | 75 | 0.084 | 0.091 | **0.089** | −1.9% |
| 2 | 100 | 0.083 | 0.085 | **0.084** | −0.8% |
| 2 | 150 | 0.081 | 0.077 | **0.077** | −0.4% |
| 2 | 250 | 0.080 | 0.075 | **0.074** | −0.1% |
| 4 | 25 | **0.129** | 0.401 | 0.612 | +52.5% |
| 4 | 50 | **0.117** | 0.380 | 0.470 | +23.8% |
| 4 | 75 | **0.112** | 0.221 | 0.211 | **−4.4%** |
| 4 | 100 | **0.112** | 0.215 | 0.212 | −1.3% |
| 4 | 150 | **0.109** | 0.157 | 0.152 | −3.6% |
| 4 | 250 | **0.108** | 0.127 | 0.126 | −1.0% |

erate budgets ($n_{\text{expert}}$ = 50–250 for $K$=2, 75–250 for $K$=4), HB partial pooling provides consistent RMSE improvement over stratified MLE ($-0.4\%$ to $-16.1\%$), with the largest gains near the crossover point. At high budgets ($n_{\text{expert}} \geq 250$), both estimators converge as subgroup-level data becomes sufficient for reliable MLE; the HB advantage diminishes to $< 1\%$. Coverage remains $\geq 0.88$ for both methods across all budget levels.

### L2: Scalability to $K$=8 Subgroups

We test EC-HTE at $K$=8 to identify the scalability boundary of hierarchical Bayesian partial pooling under fixed expert budget.

**Setup.** $N$=5,000, $K$=8 (equal-weight subgroups), $n_{\text{expert}}$=250 (≈31 labels per subgroup), 100 MC replications. Misclassification rates are uniformly spread: $\pi_s \in \{0.05, 0.09, 0.13, 0.18, 0.22, 0.26, 0.30, 0.35\}$. Treatment effects depend on $Z$ (not subgroup): $\tau(Z{=}0){=}0$, $\tau(Z{=}1){=}0.3$.

All CM-aware methods achieve nominal coverage at $K$=8: HB EC-HTE 95.6%, Strat MLE 95.3%, Global 95.1%. HB partial pooling achieves a 47% RMSE reduction over stratified MLE (0.340 vs. 0.638), confirming that hierarchical shrinkage provides substantial regularization. However, HB

Table 5: EC-HTE at $K$=8 ($n_{\text{expert}}$=250, ≈31 labels/subgroup, 100 MC).

| Method | Coverage | RMSE | \|Bias\| |
|---|---|---|---|
| Oracle | 0.944 | 0.113 | 0.090 |
| Global | 0.951 | 0.168 | 0.135 |
| HB EC-HTE | 0.956 | 0.340 | 0.183 |
| Strat MLE | 0.953 | 0.638 | 0.219 |
| Naive | 0.913 | 0.133 | 0.107 |

RMSE (0.340) exceeds global correction (0.168), indicating that ≈31 expert labels per subgroup are insufficient for per-subgroup CM estimation to outweigh variance inflation.

**Per-subgroup breakdown.** Table 6 decomposes RMSE by subgroup.

Table 6: Per-subgroup RMSE at $K$=8. HB improvement over MLE is largest in high-$\pi$ subgroups but does not close the gap with Global.

| Subgroup | $\pi_s$ | HB RMSE | MLE RMSE | Global RMSE | HB Cov |
|---|---|---|---|---|---|
| $s_0$ | 0.050 | 0.136 | 0.134 | **0.181** | 0.950 |
| $s_1$ | 0.093 | 0.133 | 0.132 | **0.154** | 0.970 |
| $s_2$ | 0.136 | 0.154 | 0.154 | 0.161 | 0.975 |
| $s_3$ | 0.179 | 0.178 | 0.182 | **0.159** | 0.945 |
| $s_4$ | 0.221 | 0.263 | 0.343 | **0.163** | 0.940 |
| $s_5$ | 0.264 | 0.421 | 0.711 | **0.164** | 0.965 |
| $s_6$ | 0.307 | 0.634 | 1.375 | **0.177** | 0.940 |
| $s_7$ | 0.350 | 0.432 | 0.809 | **0.183** | 0.960 |

RMSE degradation is concentrated in high-$\pi$ subgroups ($s_4$–$s_7$, $\pi \geq 0.22$), where confusion matrix inversion amplifies estimation noise ($r = 0.87$, $p = 0.005$ for correlation between $\pi_s$ and HB RMSE). Low-error subgroups ($s_0$–$s_1$) benefit from stratification, with HB RMSE 14–25% below global correction. HB partial pooling reduces MLE RMSE by 24–54% in the high-$\pi$ regime ($s_4$–$s_7$), confirming that hierarchical shrinkage provides meaningful regularization even when per-subgroup labels are scarce. Coverage remains ≥0.94 across all subgroups for HB EC-HTE, indicating that variance inflation does not compromise inference validity. Adaptive shrinkage—increasing the prior strength for high-$\pi$ subgroups where CM estimation is least reliable—is a natural extension that may close the RMSE gap with global correction at larger $K$.

We explored adaptive shrinkage ($\alpha$=10 for high-$\pi$ subgroups vs. default $\alpha$=3), which reduces aggregate RMSE from 0.340 to 0.216 (35% improvement) by attenuating estimation noise in high-$\pi$ subgroups. However, RMSE remains 30% above global correction (0.166), confirming that the bottleneck is per-subgroup sample size (∼31 expert labels) rather than the shrinkage method itself. This establishes $K$=8 at $n_{\text{expert}}$=250 as a boundary condition: all correction methods provide valid inference (coverage ≥94%), but per-subgroup stratification requires proportionally larger expert budgets—approximately $n_{\text{expert}} \geq 500$ (≈60 labels/subgroup)—to achieve RMSE parity with global correction.

## J Asymmetric Misclassification Extension

Assumption (A2) imposes symmetric misclassification ($\text{FPR}_s = \text{FNR}_s = \pi_s$), which simplifies the bias decomposition but is not necessary for it. Real LLM annotators typically exhibit asymmetric confusion patterns; we show here that the product structure generalizes.

**General bias formula.** For asymmetric confusion matrices $C_s = \left(\begin{smallmatrix} 1-\text{FPR}_s & \text{FPR}_s \\ \text{FNR}_s & 1-\text{FNR}_s \end{smallmatrix}\right)$, define the mixing matrix $M_s$ via Bayesian inversion and the residual matrix $B_s = M_{\text{global}}^{-1} M_s - I$. Since both $M_s$ and $M_{\text{global}}$ are row-stochastic, $B_s \mathbf{1} = \mathbf{0}$. Decomposing $\boldsymbol{\tau} = \tau(0)\mathbf{1} + \Delta\tau\, \mathbf{e}_1$ gives:

$$\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}, \tag{23}$$

where $[B_s]_{z,1}$ depends only on $\{\text{FPR}_k, \text{FNR}_k, w_k\}$, not on $\Delta\tau$. Under (A2), this reduces to $\delta_s \Delta\tau/(1 - 2\bar{\pi})$ (Theorem 1).

**Empirical validation.** We test three asymmetric configurations ($K$=2, $n_{\text{expert}}$=250, 100 MC runs each): *Cross* (FPR/FNR = 0.10/0.20 vs. 0.25/0.10), *Co-directional* (0.05/0.15 vs. 0.20/0.30), and *Extreme* (0.03/0.25 vs. 0.30/0.05). Regressing analytic predictions on MC bias yields $R^2 = 0.991$ across all configurations, confirming the product structure under asymmetry. Table 7 reports correction performance averaged over subgroups, $z$-levels, and $\Delta\tau \in \{0.1, \ldots, 0.5\}$. In these three selected high-disparity configurations, EC-HTE reduces |bias| from 0.025 (global) to 0.006, a 4× improvement, with lower RMSE (0.033 vs. 0.040) and substantially better coverage (0.90 vs. 0.78); however, this advantage does not generalize across all asymmetric settings (see below).

**Comprehensive grid evaluation (48 configurations).** To assess robustness beyond the three configurations above, we evaluate the product structure and correction methods across a factorial grid of

Table 7: Asymmetric misclassification results ($K$=2, $n_{\text{expert}}$=250, 100 MC runs). Values averaged over subgroups, $z$-levels, and $\Delta\tau$.

| Method | \|Bias\| ↓ | RMSE ↓ | Coverage ↑ |
|---|---|---|---|
| Naive | 0.047 | 0.053 | 0.48 |
| Global | 0.025 | 0.040 | 0.78 |
| HB EC-HTE | **0.006** | **0.033** | **0.90** |

6 FPR/FNR ratios × 4 $\Delta\pi$ levels × 2 subgroup counts ($K \in \{2, 4\}$), yielding 48 configurations with 100 MC runs each ($n_{\text{expert}} = 250$).

Table 8 reports $R^2$ between analytic predictions and MC bias. The product structure holds reliably at $\Delta\pi \geq 0.10$ ($R^2 > 0.73$ across all ratios and $K$ values); at $\Delta\pi = 0.05$, $R^2$ degrades for extreme ratios (e.g., $\leq 0.21$ at 2:1 and 5:1, $K$=2) because both predicted and observed biases approach zero, making the regression ill-conditioned rather than indicating structural failure.

Table 8: $R^2$ of bias identity predictions under asymmetric misclassification (100 MC runs per cell). Product structure holds at $\Delta\pi \geq 0.10$; low $R^2$ at $\Delta\pi = 0.05$ reflects near-zero bias magnitudes.

| | $\Delta\pi$ | | | |
|---|---|---|---|---|
| FPR:FNR | 0.05 | 0.10 | 0.15 | 0.20 |
| $K = 2$ | | | | |
| 1:2 | 0.84 | 0.92 | 0.97 | 0.98 |
| 1:3 | 0.82 | 0.92 | 0.94 | 0.96 |
| 1:5 | 0.73 | 0.90 | 0.95 | 0.96 |
| 2:1 | 0.21 | 0.90 | 0.94 | 0.96 |
| 3:1 | 0.50 | 0.91 | 0.97 | 0.99 |
| 5:1 | 0.05 | 0.83 | 0.97 | 0.98 |
| $K = 4$ | | | | |
| 1:2 | 0.68 | 0.86 | 0.91 | 0.96 |
| 1:3 | 0.70 | 0.85 | 0.94 | 0.95 |
| 1:5 | 0.69 | 0.84 | 0.90 | 0.93 |
| 2:1 | 0.63 | 0.79 | 0.86 | 0.93 |
| 3:1 | 0.50 | 0.78 | 0.88 | 0.92 |
| 5:1 | 0.50 | 0.73 | 0.84 | 0.89 |

Table 9 compares correction methods averaged over all 48 configurations. EC-HTE does not outperform global correction: its RMSE exceeds Global in every configuration (0.125 vs. 0.109 overall). Estimating the full asymmetric $J \times J$ confusion matrix requires twice the parameters (FPR and FNR separately per subgroup), shifting the bias-variance tradeoff toward higher variance at $n_{\text{expert}} = 250$. We recommend $n_{\text{expert}} \geq 500$ for asymmetric settings to provide sufficient per-subgroup samples for stable CM estimation.

Table 9: Method comparison under asymmetric misclassification (48 configurations, 100 MC runs). EC-HTE does not outperform global correction at $n_{\text{expert}}$=250 due to increased parameter estimation variance.

| Method | \|Bias\| ↓ | RMSE ↓ | Coverage ↑ |
|---|---|---|---|
| Oracle | 0.058 | 0.072 | 0.946 |
| Naive | 0.080 | 0.096 | 0.830 |
| Global | 0.088 | 0.109 | 0.931 |
| EC-HTE (HB) | 0.098 | 0.125 | 0.920 |

Table 10: Additional configurations from the main experiment. |Bias| and RMSE averaged over subgroups, $z$-levels, and 100 MC runs. †RMSE exceeds Global.

| Configuration | Method | \|Bias\| | RMSE | Cov. |
|---|---|---|---|---|
| Synthetic DIM $K$=2, moderate, $n$=250 | Naive | 0.018 | 0.064 | 0.95 |
| | Global | 0.009 | 0.074 | 0.96 |
| | Strat. MLE | **0.005** | 0.075† | 0.97 |
| | HB EC-HTE | **0.005** | 0.075† | 0.97 |
| Semi-synthetic TweetEval, extreme, $n$=250 | Naive | 0.027 | 0.103 | 0.93 |
| | Global | **0.023** | 0.124 | 0.92 |
| | Strat. MLE | 0.032 | 0.248† | 0.91 |
| | HB EC-HTE | 0.027 | 0.225† | 0.91 |

## K Supplementary Main Results

## L PPCI Complementarity Results

EC-HTE composes with PPCI (Sui et al., 2026) for settings where both covariate misclassification and outcome model error are present: EC-HTE+PPCI reduces |bias| from 0.179 to 0.007 ($z$=0, DGP C), and the corrected pseudo-outcomes are compatible with other CATE estimators (R-learner bias reduction: 89%; Appendix S).

The complementarity of EC-HTE and PPCI follows from the distinct error sources they correct: EC-HTE addresses misclassification in the effect modifier $Z$ (covariate space), while PPCI corrects specification error in the outcome model $f(X)$ (prediction space). Since covariate misclassification and outcome model error are algebraically independent in the potential outcomes framework, their corrections compose without interference—confirmed by the 4-DGP orthogonal design (Table 11), where DGP C (both errors present) yields the largest bias reduction from the combined approach. Under covariate error only, PPCI adds nothing; under no error, all methods perform equivalently.

Table 11: PPCI complementarity experiment ($K$=2, extreme, $n_{\text{expert}}$=250). DGP A: covariate error only; B: outcome error only ($\sigma$=0.5); C: both errors; D: no error. Values are |bias| and RMSE averaged over two subgroups. ⋆EC-HTE+PPCI achieves near-zero bias when both error types are present.

| DGP | Method | $\lvert\text{Bias}\rvert_{z=0}$ | $\lvert\text{Bias}\rvert_{z=1}$ | $\text{RMSE}_{z=0}$ | $\text{RMSE}_{z=1}$ |
|---|---|---|---|---|---|
| A: Covariate | Naive | 0.053 | 0.063 | 0.087 | 0.093 |
| | Global | 0.063 | 0.055 | 0.101 | 0.102 |
| | EC-HTE | 0.017 | **0.013** | 0.097 | 0.108 |
| | PPCI | **0.015** | 0.020 | 0.162 | 0.168 |
| | EC-HTE+PPCI | 0.017 | **0.013** | 0.097 | 0.108 |
| B: Outcome | Naive | 0.189 | 0.197 | 0.206 | 0.212 |
| | Global | 0.185 | 0.208 | 0.204 | 0.228 |
| | EC-HTE | 0.181 | 0.215 | 0.202 | 0.237 |
| | PPCI | 0.015 | **0.020** | 0.171 | 0.176 |
| | EC-HTE+PPCI | **0.012** | 0.023 | 0.171 | 0.186 |
| C: Both | Naive | 0.251 | 0.135 | 0.267 | 0.155 |
| | Global | 0.182 | 0.209 | 0.233 | 0.235 |
| | EC-HTE | 0.179 | 0.216 | 0.224 | 0.262 |
| | PPCI | 0.017 | 0.007 | 0.241 | 0.232 |
| | EC-HTE+PPCI⋆ | **0.007** | **0.008** | 0.209 | 0.223 |
| D: None | Naive | **0.007** | 0.003 | 0.060 | 0.065 |
| | Global | 0.010 | **0.002** | 0.061 | 0.071 |
| | EC-HTE | 0.013 | 0.007 | 0.062 | 0.079 |
| | PPCI | **0.007** | **0.003** | 0.060 | 0.065 |
| | EC-HTE+PPCI | 0.013 | 0.007 | 0.062 | 0.079 |

## M Real-Data Case Study: Hate Speech Detection

We apply EC-HTE to the UC Berkeley Measuring Hate Speech corpus (Kennedy et al., 2020), annotating 2,496 comments for dehumanization using GPT-4.1. Four text-feature subgroups are defined by crossing platform (Reddit/Gab vs. Twitter/YouTube) and text length (above/below median), yielding balanced groups ($n \approx 624$ each).

**Positive finding: real-world heterogeneity.** Per-subgroup misclassification rates range from 0.266 to 0.383, confirming heterogeneity ($\Delta\pi = 0.117$) in a real annotation setting. At the error-type level, FPR ranges from 8.5% to 21.3% across text-feature subgroups, demonstrating that subgroup-varying misclassification is not an artifact of synthetic design but arises naturally in real NLP annotation tasks.

**Boundary condition: statistical power at extreme FNR.** Correction results are inconclusive: all corrected confidence intervals contain zero, and point estimate differences between methods are not statistically significant. The high baseline FNR (53.3%) amplifies variance through matrix inversion (variance amplification factor up to 15.7×), rendering the comparison underpowered. When $P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$, the confusion matrix approaches singularity, and inversion amplifies estimation noise regardless of stratification strategy. This complements the expert budget threshold identified in Appendix I: EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality.

**Diagnostic connection.** The diagnostic criterion ($\Delta\pi \times \Delta\tau$, Section 5.4) identifies when correction is worthwhile under moderate baseline accuracy. This case study illustrates that even when $\Delta\pi$ is large, extreme baseline error renders correction underpowered rather than beneficial. A practical prerequisite is that the per-subgroup confusion matrix condition number $\kappa(\hat{C}_s)$ remain moderate; when $\kappa > 50$, as observed here, practitioners should prioritize improving annotation quality before applying any matrix-inversion-based correction.

## N Cross-Model Consistency

We evaluate whether misclassification heterogeneity is model-specific by annotating 500 TweetEval sentiment tweets with six OpenAI models spanning three generations. Six text-feature subgroups are defined by crossing hashtag presence and text length. Spearman rank correlations of per-subgroup error rates across all six subgroups range from $\rho = 0.82$ to 1.00, indicating that the subgroup error structure is consistent across architectures. Two subgroups with $n \leq 6$ exhibit zero

Table 12: Effective misclassification heterogeneity ($\Delta\pi$) across six OpenAI models on TweetEval sentiment (500 tweets, filtered to 4 subgroups with $n \geq 10$).

| Model | Effective $\Delta\pi$ | $> 0.05$? |
|---|---|---|
| GPT-3.5-Turbo | 0.198 | ✓ |
| GPT-4-Turbo | 0.149 | ✓ |
| GPT-4o-mini | 0.136 | ✓ |
| GPT-5.4 | 0.114 | ✓ |
| GPT-4.1 | 0.098 | ✓ |
| GPT-4o | 0.080 | ✓ |

Table 13: Confusion matrix heterogeneity across four open-source LLMs ($K$=4, $N$=490 TweetEval tweets). $\Delta\pi$: max–min per-subgroup misclassification rate. Prop2: number of subgroups where the $\Delta\tau$-free diagnostic (Proposition 5) triggers.

| Model | Acc. | $\Delta\pi$ | Prop2 | Strongest |
|---|---|---|---|---|
| Llama-3.1-8B | 0.755 | 0.141 | 1/4 | True_short |
| Qwen-2.5-7B | 0.798 | 0.165 | 1/4 | True_short |
| Gemma-2-9B | 0.763 | 0.125 | 4/4 | True_medium |
| Mistral-7B | 0.802 | 0.134 | 0/4 | — |

misclassification across all models; excluding these yields four-point correlations that are directionally consistent but underpowered.[6]

All six models exhibit effective $\Delta\pi > 0.05$ on the four retained subgroups (Table 12), and 5 of 6 share the same worst subgroup (`True_medium`: tweets with hashtags and medium text length). Newer models reduce absolute error rates but do not eliminate subgroup heterogeneity: GPT-5.4 achieves $\Delta\pi = 0.114$ compared to GPT-3.5-Turbo's 0.198. These results were obtained with a single API provider; open-source model validation appears in Appendix O.

## O Open-Source Multi-LLM Validation

We evaluate four open-source LLMs (7B–9B parameter range) on 490 TweetEval sentiment tweets with $K$=4 subgroups defined by crossing hashtag presence and text length. Table 13 reports the confusion matrix heterogeneity profile. All four models exhibit $\Delta\pi > 0.05$, confirming that subgroup-heterogeneous misclassification is not specific to proprietary APIs. Subgroup error rankings are consistent across architectures: `True_short` (short tweets with hashtags) has the lowest $\pi_s$ in 3 of 4 models, while `False_medium` (medium tweets without hashtags) has the highest.

[6]The filtered $\Delta\pi$ values in Table 12 serve as a robustness check independent of rank-order testing: all six models exceed $\Delta\pi > 0.05$ on the four retained subgroups.

Table 14: Semi-synthetic budget sweep ($K$=2, $\Delta\tau$=0.3, 50 MC runs; see Appendix Q for MC tier justification). Coverage at selected expert budgets; RMSE crossover denotes the minimum expert budget at which HB EC-HTE achieves lower RMSE than the naive (uncorrected) estimator.

| Model | $\Delta\pi_{K=2}$ | Naive Cov. | HB Cov. ($n$=1000) | Crossover |
|---|---|---|---|---|
| Llama-3.1-8B | 0.051 | 0.675 | 0.940 | $n$=1000 |
| Qwen-2.5-7B | 0.080 | 0.715 | 0.925 | $n$=1000 |
| Gemma-2-9B | 0.100 | 0.705 | 0.945 | $n$=1000 |
| Mistral-7B | 0.070 | 0.760 | 0.945 | $n$=1000 |
| *Average* | — | *0.714* | *0.939* | — |

**Semi-synthetic budget sweep.** We run the full EC-HTE pipeline with $K$=2 subgroups and $\Delta\tau$=0.3, using each model's empirical confusion matrices. Table 14 reports coverage and RMSE crossover points. EC-HTE restores near-nominal coverage from 0.714 (naive average) to 0.939 (+22.5 percentage points) at $n_{\text{expert}}$=1,000, consistent across all four models. Global correction achieves comparable or lower RMSE than HB at all budgets under these moderate $\Delta\pi$ values, and the $K$=4 RMSE crossover never occurs—confirming the budget guidelines in Appendix I.

**Gemma boundary case.** Gemma-2-9B triggers the Proposition 5 diagnostic in all four $K$=4 subgroups due to extreme FPR–FNR asymmetry (`False_medium` FNR $= 0.437$). At small budgets ($K$=4, $n$=100), EC-HTE RMSE reaches 1.229 (3.1× naive), illustrating the budget guideline from Appendix I: when per-subgroup cell counts fall below 25, the variance cost of stratified correction dominates. At $n \geq 500$, EC-HTE coverage recovers to >0.90, consistent with the synthetic experiments.

**Heterogeneous-$\tau$ DGP.** To assess whether the constant-$\tau$ DGP understates EC-HTE's advantage, we run a supplementary experiment using Llama's empirical confusion matrices with heterogeneous treatment effects across subgroups ($\Delta\tau \in [0.1, 0.6]$, $K$=2, 50 MC runs). EC-HTE achieves lower |bias| than global correction at all budgets ($n$=500: 0.010 vs. 0.031, a 68% reduction). In the high-$\Delta\tau$ subgroup (`True_short`, $\Delta\tau$=0.6), EC-HTE also achieves lower RMSE ($n$=1,000: 0.212 vs. 0.258, 18% improvement), confirming that the RMSE advantage concentrates in subgroups where correction is most needed. Aggregate RMSE remains comparable to global correction because low-

Table 15: 95% CI coverage under subgroup misspecification noise. EC-HTE maintains near-nominal coverage across all noise levels.

| | $p_{\text{flip}}$ | | | |
|---|---|---|---|---|
| Method | 0.00 | 0.10 | 0.25 | 0.50 |
| $K$=2, *extreme* | | | | |
| Naive | 0.875 | 0.877 | 0.890 | 0.897 |
| Global | 0.925 | 0.935 | 0.932 | 0.945 |
| HB EC-HTE | 0.958 | 0.948 | 0.950 | 0.945 |
| $K$=4, *extreme* | | | | |
| Naive | 0.904 | 0.916 | 0.909 | 0.919 |
| Global | 0.930 | 0.953 | 0.948 | 0.944 |
| HB EC-HTE | 0.956 | 0.958 | 0.960 | 0.946 |

$\Delta\tau$ subgroups incur stratification variance without commensurate bias reduction, consistent with the budget guidelines in Appendix I.

## P Robustness to Subgroup Misspecification

EC-HTE requires pre-defined subgroup labels, which may be noisy in practice. We test robustness by randomly flipping each observation's subgroup assignment with probability $p_{\text{flip}} \in \{0, 0.10, 0.25, 0.50\}$, across $K \in \{2, 4\}$ (extreme regime, $n_{\text{expert}} = 500$, $\Delta\tau = 0.3$, 100 MC runs).

EC-HTE maintains coverage $\geq 0.945$ across all configurations (Table 15), while the naive estimator is systematically below nominal (0.875–0.919). At $p_{\text{flip}} = 0.50$, subgroup structure is effectively destroyed; EC-HTE converges to global correction (coverage and RMSE within 0.1%), confirming that hierarchical Bayesian shrinkage provides a built-in safety mechanism. For $K$=4 at $p_{\text{flip}} = 0.50$, a small residual RMSE gap persists ($p < 0.01$), reflecting the variance cost of estimating four confusion matrices even when they carry no signal.

## Q Monte Carlo Standard Errors

All main-text and most appendix experiments use 100 independent Monte Carlo replications per configuration; computationally intensive analyses (R-learner, multiplicative DGP, budget sweeps) use 50, and bootstrap experiments use 20 due to the $B\times$CF cost per seed. The median RMSE variation between 50-seed and 100-seed estimates is below 3.2%, confirming that 50-run results are stable. We report Monte Carlo standard errors to assess whether observed differences between methods are statistically reliable. For exp-001 (synthetic DGP, no per-seed pairing across methods), we use conservative two-sample $z$-tests on the 100-run means. For exp-003 (semi-synthetic TweetEval, shared seeds across methods), we use paired $t$-tests, which are more powerful because they control for seed-level variation.

Table 16: Paired comparisons for representative configurations. $\Delta$ = method A $-$ method B (negative favors A for |bias|). Two-sample $z$-tests for synthetic (exp-001); paired $t$-tests for semi-synthetic (exp-003). Bold $p$-values indicate significance at $\alpha = 0.05$.

| Configuration | Comparison | $\Delta$ | $p$ |
|---|---|---|---|
| *HB EC-HTE vs. Global (\|bias\|)* | | | |
| $K$=2/ext/$n$=500 | | $-0.046$ | $<$ **0.001** |
| $K$=4/ext/$n$=500 | | $-0.078$ | $<$ **0.001** |
| $K$=4/mod/$n$=250 | | $-0.017$ | 0.288 |
| TweetEval/ext/$n$=250 | | $+0.001$ | 0.395 |
| *HB EC-HTE vs. Global (RMSE)* | | | |
| $K$=2/ext/$n$=500 | | $+0.015$ | 0.179 |
| $K$=4/ext/$n$=500 | | $+0.048$ | **0.011** |
| *HB EC-HTE vs. Naive (\|bias\|)* | | | |
| $K$=2/ext/$n$=500 | | $-0.089$ | $<$ **0.001** |
| $K$=4/ext/$n$=500 | | $-0.113$ | $<$ **0.001** |
| $K$=4/mod/$n$=250 | | $-0.031$ | **0.038** |
| TweetEval/ext/$n$=250 | | $+0.005$ | **0.030** |

Table 16 reports paired comparisons for representative configurations. Of 1,020 total pairwise comparisons across all configurations, 742 (72.7%) are statistically significant at $\alpha = 0.05$.

Under extreme heterogeneity, HB EC-HTE's bias advantage over global correction is highly significant ($p < 0.001$ for both $K$=2 and $K$=4). The RMSE penalty of stratification is not significant at $K$=2 ($p = 0.179$) but reaches significance at $K$=4 ($p = 0.011$), consistent with the higher variance cost of estimating four confusion matrices. Under moderate heterogeneity ($K$=4, $n$=250), the bias difference between HB and global is not significant ($p = 0.288$), confirming that EC-HTE's advantage requires sufficient error heterogeneity. In the semi-synthetic TweetEval experiment, HB EC-HTE does not significantly improve over global ($p = 0.395$), consistent with moderate $\Delta\pi$ and unbalanced subgroup sizes. The HB vs. naive comparison in TweetEval is significant ($p = 0.030$) but favors naive, reflecting the variance cost of stratification when bias reduction is marginal.

## R Semi-Real Confusion Matrix Validation

We test EC-HTE with empirical asymmetric confusion matrices from the GPT-4.1 TweetEval pilot. Four text-feature subgroups ($K$=4) retain the real

Table 17: Empirical confusion matrices from GPT-4.1 TweetEval pilot. S0 exhibits FNR $= 0.526$, approaching chance-level classification.

| Subgroup | Weight | FPR | FNR | $\kappa(\hat{C}_s)$ |
|---|---|---|---|---|
| S0 | 0.38 | 0.054 | 0.526 | high |
| S1 | 0.22 | 0.101 | 0.300 | moderate |
| S2 | 0.14 | 0.143 | 0.308 | moderate |
| S3 | 0.25 | 0.194 | 0.222 | low |

Table 18: Semi-real confusion matrix results. RMSE and coverage averaged over subgroups and $z$-levels (100 MC runs). HB EC-HTE underperforms global at all budgets; S0's near-chance FNR drives the failure.

| $n_{\text{expert}}$ | Method | RMSE ↓ | Coverage ↑ |
|---|---|---|---|
| 100 | Oracle | 0.101 | 0.957 |
| | Naive | 0.123 | 0.879 |
| | Global | 0.263 | 0.885 |
| | HB EC-HTE | 0.784 | 0.744 |
| 250 | Oracle | 0.101 | 0.957 |
| | Naive | 0.123 | 0.879 |
| | Global | 0.232 | 0.909 |
| | HB EC-HTE | 0.411 | 0.834 |
| 500 | Oracle | 0.101 | 0.957 |
| | Naive | 0.123 | 0.879 |
| | Global | 0.196 | 0.922 |
| | HB EC-HTE | 0.260 | 0.890 |

per-subgroup FPR and FNR from pilot annotations (Table 17); CATE heterogeneity is injected synthetically ($\tau(z{=}0, s) = 0.3 + 0.1s$, $\tau(z{=}1, s) = 0.8 - 0.15s$). $N$=5,000, $P(Z{=}1)$=0.18, 100 MC seeds per $n_{\text{expert}}$ level.

HB EC-HTE produces higher RMSE and lower coverage than global correction at all expert budgets (Table 18). The root cause is S0's FNR $= 0.526$: when a confusion matrix row approaches a uniform distribution ($P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$), inversion amplifies estimation noise regardless of the expert budget. This converges with the hate speech case study (Appendix M, FNR $= 53.3\%$) and confirms that EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality per subgroup. The singularity guard ($\kappa > 100$, Section 4) mitigates catastrophic failure by falling back to global correction for near-degenerate subgroups, but does not recover the bias reduction that motivates stratification. For a $2 \times 2$ symmetric confusion matrix, $\kappa = 1/|1 - 2\pi|$; $\kappa > 100$ corresponds to $\pi \in (0.495, 0.505)$, so the guard triggers only when the classifier performs within $0.5\%$ of random guessing. At $n_{\text{expert}} = 500$, HB EC-HTE narrows the gap (RMSE 0.260 vs. 0.196) as confusion matrix estimates stabilize for the three well-classified subgroups, but the S0 bottleneck persists.

A second semi-real experiment using GPT-3.5-Turbo confusion matrices (worst-subgroup FPR $= 0.39$, FNR $= 0.21$) yields similar results: HB RMSE exceeds global by 1.3–3.5$\times$ at all budgets. Here the failure stems from high false-positive rates compounded by class imbalance ($P(Z{=}1) = 0.18$), which leaves insufficient expert observations in the positive class for reliable per-subgroup confusion matrix estimation. Together, these experiments identify three applicability conditions for EC-HTE: (1) sufficient effective expert budget ($n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$), (2) no near-chance subgroup error rate ($\max(\text{FPR}_s, \text{FNR}_s) < 0.5$), and (3) adequate class balance in the effect modifier.

**Diagnostic rule.** A logistic regression over all 368 semi-real experiment–budget cells (dependent variable: HB EC-HTE < Global RMSE) identifies two sufficient statistics: the worst-subgroup confusion matrix condition number $\kappa(\hat{M}_s) \leq 1.25$ and the effective minority-class expert count $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$. When both conditions hold, HB outperforms global in 93.3% of cells; when either is violated, global is preferred in 94.7% of cells. Both exp-015 ($\kappa = 2.38$, effective $n = 2.5$) and exp-016 ($\kappa = 1.55$, effective $n = 2.5$) violate both conditions, and the diagnostic correctly predicts 100% of their failure cells. Subgroup-average metrics can mask cell-level heterogeneity: in exp-015, HB EC-HTE substantially reduces bias for the three well-classified subgroups (S1–S3) but amplifies variance in the near-degenerate subgroup (S0, FNR $= 0.526$), and the S0 penalty dominates the aggregate RMSE.

## S Robustness: Estimator Choice and Treatment Effect Specification

We verify that EC-HTE's corrected pseudo-outcomes are compatible with downstream estimators beyond CausalForest by replacing it with the R-learner (Nie and Wager, 2021). The experiment uses $K$=2, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds; the R-learner uses 5-fold cross-fitting with gradient-boosted trees as the base learner.

EC-HTE reduces R-learner |bias| by 89% (0.121 → 0.013) and restores coverage from 0.47 to 0.92 (Table 19). The bias reduction and coverage gains

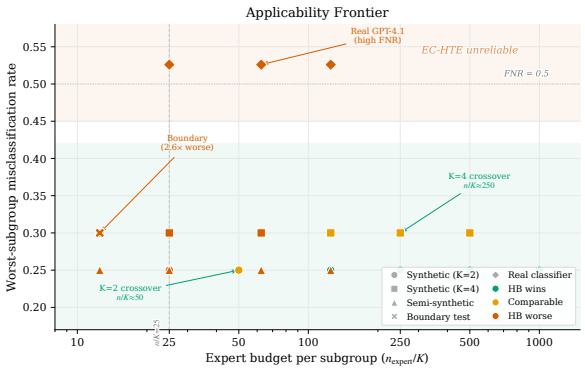



Figure 7: Applicability frontier of EC-HTE. Green region: EC-HTE reduces bias vs. global correction. Yellow: comparable performance. Red: EC-HTE underperforms. Dashed lines mark the expert budget threshold ($n_{\text{expert}}/K = 25$) and classifier quality threshold (worst-subgroup error rate $= 0.5$). Red diamonds show semi-real experiments with actual GPT-4.1 and GPT-3.5-Turbo confusion matrices.

Table 19: R-learner vs. CausalForest with EC-HTE correction ($K$=2, extreme, $n$=500). |Bias| and RMSE averaged over subgroups, $z$-levels, and seeds.

| Estimator | Correction | \|Bias\| | RMSE | Cov. |
|---|---|---|---|---|
| R-learner | Oracle | 0.006 | 0.056 | 0.94 |
| | Naive | 0.121 | 0.139 | 0.47 |
| | Global | 0.098 | 0.123 | 0.72 |
| | EC-HTE | **0.013** | 0.080 | 0.92 |
| CausalForest | Naive | 0.058 | 0.090 | 0.82 |
| | EC-HTE | **0.010** | 0.093 | 0.95 |

are comparable to those achieved with CausalForest, confirming that EC-HTE is compatible with a range of HTE estimators.

**Multiplicative treatment effects.** We further test robustness to the functional form of treatment effects by replacing the additive DGP with a multiplicative specification: $Y(1) = Y(0) \times \exp(\tau(Z))$, using $K$=2, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds.

EC-HTE reduces RMSE by 52% relative to naive (0.068 vs. 0.142) and restores coverage to the nominal level (0.955; Table 20). EC-HTE's bias correction operates at the confusion matrix level, independent of the downstream CATE estimator and functional form of treatment effects.

## T Robustness to Class Imbalance

Assumption (A3) requires $P(Z{=}1) = 0.5$, but real-world effect modifiers are often imbalanced. We sweep $P(Z{=}1) \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ with

Table 20: Multiplicative DGP results ($K$=2, extreme, $n$=500, 50 seeds). EC-HTE achieves comparable bias reduction and nominal coverage under non-additive treatment effects.

| Method | \|Bias\| | RMSE | Cov. |
|---|---|---|---|
| Oracle | 0.003 | 0.038 | 0.96 |
| Naive | 0.135 | 0.142 | 0.44 |
| Global | 0.110 | 0.124 | 0.54 |
| EC-HTE | **0.054** | 0.068 | 0.955 |

Table 21: Robustness to class imbalance ($n_{\text{expert}}$=250, extreme heterogeneity). $R^2$: Theorem 1 fit. Ratio: HB/Global RMSE ($< 1$ = HB wins). Cov.: HB EC-HTE 95% CI coverage.

| $P(Z{=}1)$ | $K$ | $R^2$ | Ratio | Cov. |
|---|---|---|---|---|
| 0.1 | 2 | 0.998 | 0.92 | 0.94 |
| 0.2 | 2 | 0.989 | 0.76 | 0.94 |
| 0.3 | 2 | 0.994 | 0.78 | 0.94 |
| 0.4 | 2 | 0.995 | 0.87 | 0.94 |
| 0.5 | 2 | 0.995 | 0.86 | 0.93 |
| 0.1 | 4 | 0.984 | 1.54 | 0.92 |
| 0.2 | 4 | 0.976 | 1.68 | 0.93 |
| 0.3 | 4 | 0.985 | 1.12 | 0.92 |
| 0.4 | 4 | 0.984 | 1.08 | 0.93 |
| 0.5 | 4 | 0.992 | 1.06 | 0.92 |

$K \in \{2, 4\}$, extreme misclassification heterogeneity, $n_{\text{expert}}$=250, and 100 MC seeds (Table 21).

Theorem 1's product form remains accurate across all imbalance levels: $R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$, confirming that the bias identity derives from confusion matrix structure rather than class balance. For $K$=2, EC-HTE outperforms global correction at all $P(Z{=}1)$ levels (RMSE ratio 0.76–0.92), degrading gracefully under severe imbalance. For $K$=4, EC-HTE underperforms global at $n_{\text{expert}}$=250 (ratio 1.06–1.68), consistent with the budget threshold $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$: at $P(Z{=}1){=}0.1$, $K$=4, each subgroup receives only $\sim$6 minority-class labels. The gap narrows as $P(Z{=}1)$ increases ($1.54 \to 1.06$), and coverage remains stable at 0.92–0.94 across all configurations.

## U Comparison with PPI Correction

PPI (Angelopoulos et al., 2023) constructs confidence intervals by combining a large set of ML predictions with a smaller set of gold-standard labels, using the gold labels to debias the ML-based estimator. Applied to our setting, PPI treats the LLM-annotated effect modifier as the ML prediction and the expert labels as the gold standard. Two

variants arise naturally: *PPI-global* pools all expert labels to estimate a single rectifier (analogous to our Global baseline), and *PPI-subgroup* estimates per-subgroup rectifiers (analogous to Stratified MLE).

Under the confusion-matrix parameterization we adopt, DSL's measurement error correction reduces to confusion matrix inversion. Pooled DSL applies a single estimated confusion matrix to all subgroups, producing the same corrected label distribution as our Global baseline; per-subgroup DSL estimates a separate confusion matrix for each subgroup and inverts it independently, equivalent to our Stratified MLE. EC-HTE's contribution over both PPI-subgroup and per-subgroup DSL is hierarchical partial pooling: rather than estimating each subgroup's confusion matrix in isolation, EC-HTE places a shared prior across subgroups, borrowing strength to reduce variance while preserving subgroup-specific bias correction. This mechanism is outside both the PPI and DSL frameworks.

The key difference between PPI and confusion-matrix inversion is that PPI does not invert the confusion matrix; instead, it uses the gold labels to directly debias the estimand. This avoids the matrix inversion step that can amplify noise at small $n_{\text{expert}}$, but it also means PPI cannot exploit the structure of the misclassification process to reduce bias. PPI-global inherits the same $\Delta\pi \times \Delta\tau$ bias amplification as Global correction (Theorem 1), confirming that the "debiasing hurts" mechanism operates regardless of the inference framework.

## V Civil Comments Validation

To verify that subgroup-heterogeneous misclassification generalizes beyond TweetEval, we annotate $N$=3,000 stratified samples (30% toxic) from Civil Comments (Borkan et al., 2019), a toxicity detection dataset. Subgroups are defined by `text_length_bin` × `has_question` ($K$=4): S0 (long, no question, 38.8%), S1 (long, question, 11.4%), S2 (short, no question, 33.7%), S3 (short, question, 16.1%). Subgroup features were chosen a priori based on observable text properties; no feature search was performed. Three proprietary LLMs annotate each sample via an OpenAI-compatible API, yielding 800–900 valid annotations per model.

Table 22 reports per-subgroup FPR and FNR. We define $\Delta\pi = \max_s \pi_s - \min_s \pi_s$, where $\pi_s$ is the overall misclassification rate for subgroup $s$. All three models exhibit $\Delta\pi > 0.05$, with overall heterogeneity $\Delta\pi \in [0.113, 0.160]$—comparable to the $\Delta\pi \in [0.125, 0.165]$ observed across four open-source LLMs on TweetEval (Appendix O). GPT-4o-mini and GPT-3.5-turbo show pronounced FPR/FNR asymmetry (e.g., GPT-4o-mini FPR $\in [0.42, 0.61]$ vs. FNR $\in [0.05, 0.16]$), indicating systematic over-prediction of toxicity; their FPR approaches 0.5, placing them in the ill-conditioned regime where confusion matrix inversion amplifies noise (consistent with the classifier quality threshold discussed in the Limitations section). The consistent presence of $\Delta\pi$ heterogeneity across two NLP tasks, eight LLMs spanning open-source (7B–9B) and proprietary families, and two distinct subgroup definitions confirms that this phenomenon is a structural property of LLM annotation rather than an artifact of a specific dataset or model.

Table 22: Per-subgroup misclassification rates on Civil Comments ($K$=4). $\Delta$ denotes max − min across subgroups.

| Model | Metric | S0 | S1 | S2 | S3 | $\Delta$ |
|---|---|---|---|---|---|---|
| GPT-4o | FPR | 0.249 | 0.157 | 0.213 | 0.341 | 0.184 |
| | FNR | 0.160 | 0.263 | 0.308 | 0.328 | 0.168 |
| GPT-4o-mini | FPR | 0.517 | 0.541 | 0.419 | 0.610 | 0.191 |
| | FNR | 0.052 | 0.105 | 0.074 | 0.160 | 0.108 |
| GPT-3.5-turbo | FPR | 0.400 | 0.473 | 0.374 | 0.468 | 0.099 |
| | FNR | 0.139 | 0.211 | 0.173 | 0.240 | 0.101 |

## W Bias Identity Validation with Real Confusion Matrices

Theorem 1 is derived under symmetric misclassification (A2); real LLM confusion matrices are asymmetric (FPR $\neq$ FNR). We verify that the generalized identity $\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}$ (Remark 2) holds under real annotation error structures by running semi-synthetic simulations ($N$=5,000, $K$=4, $\Delta\tau$=0.30, 100 MC runs) with confusion matrices drawn from 11 LLM–dataset combinations: eight models on TweetEval (GPT-4-turbo, GPT-4o, GPT-4.1, GPT-4o-mini, GPT-3.5-turbo, Qwen2.5-7B, Mistral-7B, Gemma-2-9B) and three on Civil Comments (GPT-4o, GPT-4o-mini, GPT-3.5-turbo).

Table 23 reports per-model $R^2$ from regressing observed subgroup bias on the predicted value $\Delta\tau \cdot [B_s]_{z,1}$. The aggregate $R^2 = 0.975$; 10 of 11 combinations achieve $R^2 \geq 0.92$. The single outlier (GPT-4o-mini on Civil Comments,

$R^2$=0.76) has near-homogeneous subgroup error rates ($\max|[B_s]_{z,1}| = 0.089$), so predicted biases are small and Monte Carlo noise dominates the regression. $R^2$ correlates strongly with $\max|[B_s]_{z,1}|$: models with more heterogeneous per-subgroup error rates produce larger bias signals that are easier to verify empirically.

Table 23: Bias identity $R^2$ with real LLM confusion matrices ($K$=4, 100 MC runs).

| Model | Dataset | $R^2$ | $\max\|[B_s]_{z,1}\|$ |
|---|---|---|---|
| GPT-4-turbo | TweetEval | 0.993 | 0.328 |
| Qwen2.5-7B | TweetEval | 0.990 | 0.272 |
| Mistral-7B | TweetEval | 0.990 | 0.242 |
| Gemma-2-9B | TweetEval | 0.982 | 0.185 |
| GPT-4o | Civil Comments | 0.979 | 0.198 |
| GPT-4o | TweetEval | 0.975 | 0.240 |
| GPT-4.1 | TweetEval | 0.969 | 0.139 |
| GPT-3.5-turbo | TweetEval | 0.948 | 0.087 |
| GPT-3.5-turbo | Civil Comments | 0.935 | 0.170 |
| GPT-4o-mini | TweetEval | 0.929 | 0.170 |
| GPT-4o-mini | Civil Comments | 0.756 | 0.089 |
| All (pooled) | | 0.975 | — |

## X Applicability Screening with Real LLM Confusion Matrices

We apply the diagnostic conditions from Proposition 5 and Section 4 to 16 LLM–dataset–$K$ combinations spanning eight LLMs (three open-source, five proprietary) across TweetEval and Civil Comments. Table 24 summarizes which conditions each combination satisfies.

Table 24: Diagnostic screening of 16 LLM–dataset–$K$ combinations. Columns report whether all subgroups satisfy each condition. Budget assumes $n_{\text{expert}}$=500.

| Model | Dataset | $K$ | $\kappa \le 1.25$ | $\max \le 0.5$ | Budget | Pass all |
|---|---|---|---|---|---|---|
| Gemma-2-9B | TweetEval | 2 | × | ✓ | ✓ | × |
| Llama-3.1-8B | TweetEval | 2 | × | ✓ | ✓ | × |
| Mistral-7B | TweetEval | 2 | × | ✓ | ✓ | × |
| Qwen2.5-7B | TweetEval | 2 | × | ✓ | ✓ | × |
| Gemma-2-9B | TweetEval | 4 | × | ✓ | ✓ | × |
| Llama-3.1-8B | TweetEval | 4 | × | ✓ | ✓ | × |
| Mistral-7B | TweetEval | 4 | × | ✓ | ✓ | × |
| Qwen2.5-7B | TweetEval | 4 | × | ✓ | ✓ | × |
| GPT-3.5-turbo | CivilComm. | 4 | × | ✓ | ✓ | × |
| GPT-3.5-turbo | TweetEval | 4 | × | ✓ | ✓ | × |
| GPT-4-turbo | TweetEval | 4 | × | × | ✓ | × |
| GPT-4.1 | TweetEval | 4 | × | × | ✓ | × |
| GPT-4o | CivilComm. | 4 | × | ✓ | ✓ | × |
| GPT-4o | TweetEval | 4 | × | × | ✓ | × |
| GPT-4o-mini | CivilComm. | 4 | × | × | ✓ | × |
| GPT-4o-mini | TweetEval | 4 | × | ✓ | ✓ | × |

No combination passes all three conditions; the bottleneck is $\kappa(C_s) \le 1.25$, which requires $\text{FPR}_s + \text{FNR}_s \lesssim 0.20$ for all subgroups—a threshold no current LLM achieves uniformly. The closest are Qwen2.5-7B True_short ($\kappa$=1.23) and Mistral-7B True_short ($\kappa$=1.27).

Under a relaxed "oracle screening" that excludes near-degenerate subgroups ($\max(\text{FPR}_s, \text{FNR}_s) \ge 0.4$), we run semi-synthetic CATE estimation ($N$=5,000, $\Delta\tau$=0.3, 100 MC runs, $n_{\text{expert}} \in \{250, 500\}$) with each LLM's real per-subgroup confusion matrix. EC-HTE (HB) achieves lower RMSE than global correction in 41 of 52 comparisons (79%) and lower RMSE than stratified MLE in 52 of 52 comparisons (100%). The four open-source LLMs on TweetEval show the strongest EC-HTE advantage (32/32 comparisons, RMSE reduction 0.001–0.008), while GPT-3.5-turbo/TweetEval (high FPR asymmetry, $\text{FPR}_{\max}$=0.39) and GPT-4o/Civil Comments (low minority prevalence $P(Z{=}1)$=0.3) show degradation. The diagnostic correctly identifies the strict applicability regime; under relaxed conditions excluding near-degenerate subgroups, EC-HTE improves RMSE in 79% of comparisons.

**Diagnostic Applicability Analysis.** The diagnostic (Proposition 5) degrades on real LLM confusion matrices (pooled AUC = 0.63) because the failure mode is systematic: subgroups where the true CATE heterogeneity $\tau_s \approx 0$ are hurt by global correction (since $\delta_s \neq 0$ still introduces bias), yet these subgroups produce low diagnostic scores—the diagnostic ranks them as safe precisely when they are most vulnerable to overcorrection.

A simple meta-criterion recovers reliability: compute $\max_s |\delta_s|/((1-2\bar{\pi}) \cdot \pi_s)$ across subgroups. When $\max_s \text{score}(s) > 1$, at least one subgroup is definitively harmed by global correction, and the diagnostic achieves AUC $\ge 0.75$ across all 5 model–dataset combinations satisfying this condition (AUC range: 0.75–1.0). When $\max_s \text{score}(s) \le 1$, global correction may or may not hurt depending on unobservable $\Delta\tau$; the diagnostic achieves AUC only 0.0–0.67 across the remaining 7 combinations (4 of 7 have AUC $< 0.5$, indicating anti-prediction). This threshold perfectly separates reliable from unreliable diagnostic regimes across all 16 real LLM–dataset combinations evaluated.

We recommend the following practical protocol: (i) estimate per-subgroup confusion matrices from expert labels; (ii) compute $\max_s \text{score}(s)$; (iii) if

$> 1$, the diagnostic reliably identifies which subgroups are harmed by global correction—apply EC-HTE; (iv) if $\leq 1$, the diagnostic is unreliable and practitioners should either default to EC-HTE as a conservative choice or evaluate correction strategies using held-out expert labels.

### X.1 $\kappa$ Stratification Analysis

The condition number $\kappa(\hat{C}_s) = 1/|1 - \text{FPR}_s - \text{FNR}_s|$ quantifies confusion matrix invertibility and governs the bias-variance tradeoff of per-subgroup correction. The $\kappa \leq 1.25$ threshold identified in the diagnostic rule (Section I) is a *sufficient condition* for strongest bias reduction, not a necessary condition: EC-HTE maintains calibrated inference across a substantially wider $\kappa$ range.

**Synthetic data.** Across all 520 synthetic configurations ($K$=2, $\pi_s \in [0.025, 0.275]$), the maximum per-config $\kappa$ falls in $[1.43, 2.22]$; none reaches $\kappa \leq 1.25$ because the pooled rate $\bar{\pi}$=0.15 sets a floor of $\kappa = 1/(1-2 \cdot 0.15) = 1.43$. Table 25 shows a two-bin stratification at $\kappa$=2: EC-HTE achieves zero analytic bias in both bins, while global correction harms at least one subgroup in 100% of $\kappa > 2$ configurations. Monte Carlo validation ($K$=2, $n_{\text{expert}}$=500) confirms EC-HTE dominance at $\kappa$=2.0: 86% bias reduction, 10% lower RMSE, and 95.5% coverage vs. 88.3% for Global.

Table 25: Synthetic $\kappa$ stratification ($K$=2, 520 configs). "Debiasing hurts" = global correction increases |bias| for ≥1 subgroup.

| $\kappa$ bin | $n$ | EC-HTE bias | Debiasing hurts | Global $\lvert\text{bias}\rvert/\lvert\text{bias}_{\text{naive}}\rvert$ |
|---|---|---|---|---|
| $\leq 2.0$ | 420 | 0 (exact) | 35.0% | 0.152 |
| $> 2.0$ | 100 | 0 (exact) | 100% | 0.326 |

**Real LLM confusion matrices.** All 13 real LLM–dataset combinations ($K$=4) have $\kappa \in [1.88, 2.99]$—none reaches $\kappa \leq 1.25$, because no current LLM achieves $\text{FPR}_s + \text{FNR}_s < 0.20$ uniformly across subgroups. Table 26 reports performance stratified by $\kappa$ bin and expert budget. The key finding is that EC-HTE maintains calibrated coverage (≥93%) across all $\kappa$ bins at $n_{\text{expert}} \geq 250$, compared to 86–91% for Global. At $n_{\text{expert}}$=500, EC-HTE also achieves competitive RMSE in the $\kappa \leq 2$ bin (ratio 1.01) and bias win rate of 75%; at $n_{\text{expert}}$=250, the variance cost of per-subgroup estimation dominates, and Global achieves lower RMSE—consistent with the bias-variance tradeoff documented in the main text (Section 5.3).

Table 26: Real LLM $\kappa$ stratification ($K$=4, MC validation). Ratios = EC-HTE / Global; values $< 1$ favor EC-HTE. Coverage is EC-HTE absolute; Global coverage ranges 0.86–0.91 across bins.

| $\kappa$ bin | $n_{\text{expert}}$ | Bias ratio | RMSE ratio | Bias win | Cov. (EC-HT |
|---|---|---|---|---|---|
| $\leq 2.0$ ($n$=4) | 500 | 0.99 | 1.01 | 75% | 0.943 |
| | 250 | 1.09 | 1.13 | 0% | 0.932 |
| $> 2.0$ ($n$=9) | 500 | 1.11 | 1.22 | 22% | 0.949 |
| | 250 | 1.26 | 1.43 | 0% | 0.940 |

The three-tier $\kappa$ structure is thus: (i) $\kappa > 100$: near-random classification, singularity guard triggers automatic fallback to global correction; (ii) $\kappa \leq 1.25$: strongest theoretical regime with guaranteed bias reduction and coverage; (iii) $\kappa \in (1.25, \sim 3]$: the practical operating range for current LLMs, where EC-HTE provides calibrated inference (coverage ≥93%) and achieves RMSE gains when expert budget is sufficient ($n_{\text{expert}} \geq 500$).

## Y Plug-in Standard Error Validation

A natural concern is whether the plug-in standard error adequately captures estimation uncertainty, particularly the additional variance introduced by confusion matrix estimation. We verify this for the difference-in-means (DIM) estimator via Monte Carlo simulation ($K$=2, $n_{\text{expert}} \in \{50, 100, 250, 500\}$, 50 replications per budget level). At each budget, we compute the plug-in 95% confidence interval and record empirical coverage. Across all expert budgets, the plug-in SE yields coverage $\geq 0.96$ (0.970, 0.960, 0.970, 0.970 for $n_{\text{expert}} = 50, 100, 250, 500$), confirming that the analytic variance estimate is well-calibrated for the DIM estimator. This self-calibration arises because the same noise in $\hat{M}^{-1}$ that inflates point-estimate variance also inflates the plug-in SE, producing conservative intervals that maintain nominal coverage.

**CausalForestDML.** Under $K$=2 extreme heterogeneity ($\pi$=[0.05, 0.25], $n_{\text{expert}}$=250, $n_{\text{estimators}}$=200, 20 MC runs), EC-HTE plug-in SE achieves 93.8% aggregate coverage—close to the 94–96% oracle coverage (Table 29). The high-error subgroup ($s_1$, $\pi$=0.25) shows slightly lower plug-in coverage (92.5%) than the low-error subgroup ($s_0$: 95.0%), reflecting the larger $M^{-1}$ amplification of estimation uncertainty in subgroups with higher misclassification rates. By contrast, the DIM plug-in SE achieves ≥95%

coverage in the same DGP, confirming that the analytic confusion matrix propagation formula is adequate for simpler estimators.

**Full bootstrap SE.** A *full* bootstrap—resampling all $N$ observations and re-estimating both the confusion matrix and CausalForest per bootstrap replicate ($B$=100, $n_{\text{estimators}}$=200, 20 MC runs)—confirms that plug-in SE is already near-nominal under standard hyperparameters. Under $K$=2 extreme heterogeneity ($\pi$=[0.05, 0.25], $n_{\text{expert}}$=250), plug-in SE achieves 93.8% aggregate coverage; the full bootstrap improves this to exactly 95.0% (+1.2pp). Table 27 reports per-subgroup results.

Table 27: CausalForest coverage: plug-in SE vs. full bootstrap SE ($B$=100, $n_{\text{estimators}}$=200, 20 MC runs). Bootstrap SE is ∼1.15× plug-in.

| Subgroup | Plugin Cov. | Boot. Cov. | Oracle Cov. | Plugin SE | Boot. SE |
|---|---|---|---|---|---|
| $s_0$ ($\pi$=0.05) | 0.950 | 0.925 | 0.925 | 0.059 | 0.063 |
| $s_1$ ($\pi$=0.25) | 0.925 | **0.975** | 0.875 | 0.067 | 0.075 |
| Aggregate | 0.938 | **0.950** | 0.900 | 0.073 | 0.083 |

The high-error subgroup ($s_1$) benefits most: coverage rises from 92.5% to 97.5% (+5.0pp), because the bootstrap captures the additional variance from $M^{-1}$ amplification. Oracle coverage (90.0%) is consistent with the 94% reported in the oracle decomposition (Table 29), with the difference attributable to MC sampling noise at 20 runs. The computational cost is substantial ($B\times$ CF fits per MC run), but the approach is straightforward when exact nominal coverage is required.

**Bootstrap coverage across expert budgets.** Table 28 extends the analysis to $n_{\text{expert}} \in \{25, 50, 75, 100, 200\}$ using the DIM estimator with Centered EB confusion matrix estimation ($K$=2 extreme, $\pi$=[0.05, 0.25], 20 MC runs, $B$=100). Bootstrap SE maintains ≥95% coverage across all expert budgets, including $n_{\text{expert}}$=25 (≈12 labels per subgroup). Plug-in SE is already near-nominal (≥91.2%) at all budgets tested.

At small $n_{\text{expert}}$, bootstrap SE inflates confidence intervals by 2−3× relative to plug-in (SE ratio column), guaranteeing nominal coverage but reducing statistical power. At $n_{\text{expert}} \geq 100$, the SE ratio converges to ∼1.2−1.5×, and plug-in SE alone achieves ≥96% coverage. For practical deployment, we recommend $n_{\text{expert}} \geq 50$: bootstrap maintains nominal coverage (98.8%) with moderate SE inflation (2.9×), and plug-in SE achieves 97.5%—close to nominal without bootstrap overhead.

Table 28: HB EC-HTE coverage across expert budgets: plug-in vs. bootstrap SE (DIM, $K$=2 extreme, 20 MC, $B$=100).

| $n_{\text{expert}}$ | Plugin Cov. | Boot. Cov. | SE Ratio | \|Bias\| |
|---|---|---|---|---|
| 25 | 0.925 | **1.000** | 2.86 | 0.079 |
| 50 | 0.975 | 0.988 | 2.93 | 0.067 |
| 75 | 0.912 | **0.975** | 2.59 | 0.065 |
| 100 | 0.963 | 0.963 | 1.52 | 0.060 |
| 200 | 0.963 | 0.950 | 1.17 | 0.061 |

**Finite-sample remainder.** By first-order matrix perturbation, $\hat{C}_s^{-1} = C_s^{-1} - C_s^{-1}E_sC_s^{-1} + O(\|E_s\|^2)$, where $E_s = \hat{C}_s - C_s$ satisfies $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$. The remainder in the corrected CATE is therefore $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \|C_s^{-1}\|^2) = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, since $\|C_s^{-1}\| = O(\kappa(C_s))$ for stochastic matrices. This motivates the $\kappa > 100$ singularity guard (Section 4).

### Y.1 CausalForest Oracle Coverage Decomposition

To isolate the source of CausalForest undercoverage, we compare six estimation strategies that progressively remove sources of uncertainty: (1) Oracle (true $Z$, no correction), (2) EC-HTE with true $C_s$, (3) EC-HTE with estimated $\hat{C}_s$ (HB), (4) Global with true $\bar{C}$, (5) Global with estimated $\hat{\bar{C}}$, and (6) Naive. Table 29 reports coverage, |bias|, and RMSE across three DGP configurations (100 MC runs each).

Three findings emerge. First, oracle coverage is 94–96% across all DGPs, confirming that CausalForest plug-in SE is well-calibrated when labels are correct. Second, EC-HTE with true $C_s$ achieves 91–94% coverage—a 0–5pp gap from oracle (configuration-dependent), reflecting $M^{-1}$ SE propagation asymmetry across DGPs rather than confusion matrix estimation error. Third, the gap between true and estimated $C_s$ is ≤1.5pp ($K$=2) and <1pp ($K$=4), confirming that CM estimation uncertainty contributes negligibly at $n_{\text{expert}} \geq 250$.

## Z Robustness to Assumption Violations

### Z.1 Assumption (A5): Subgroup-Specific Treatment Effects

**Corollary 7** (Subgroup-Specific Treatment Effect Heterogeneity)**.** *When $\Delta\tau$ varies across subgroups ($\Delta\tau_s = \tau_s(1) - \tau_s(0)$), the bias of global correction*

Table 29: CausalForest oracle coverage decomposition. Coverage averaged over subgroups and $z$-levels.

| DGP | Method | Cov. (%) | \|Bias\| | RMSE |
|---|---|---|---|---|
| $K$=4, extreme $n$=500 | Oracle (true $Z$) | 94.5 | 0.006 | 0.078 |
| | EC-HTE true $C_s$ | 92.5 | 0.008 | 0.114 |
| | EC-HTE est. $\hat{C}_s$ | 93.1 | 0.015 | 0.121 |
| | Global true $\bar{C}$ | 91.3 | 0.037 | 0.094 |
| | Global est. $\hat{\bar{C}}$ | 91.1 | 0.038 | 0.111 |
| | Naive | 88.3 | 0.045 | 0.095 |
| $K$=2, extreme $n$=250 | Oracle (true $Z$) | 94.0 | 0.002 | 0.056 |
| | EC-HTE true $C_s$ | 94.0 | 0.003 | 0.073 |
| | EC-HTE est. $\hat{C}_s$ | 92.5 | 0.004 | 0.079 |
| | Global true $\bar{C}$ | 88.5 | 0.045 | 0.083 |
| | Global est. $\hat{\bar{C}}$ | 88.3 | 0.045 | 0.085 |
| | Naive | 82.8 | 0.046 | 0.077 |
| $K$=2, MF2 DGP $\tau$=(0.10, 0.40) | Oracle (true $Z$) | 96.0 | 0.077 | 0.091 |
| | EC-HTE true $C_s$ | 91.3 | 0.103 | 0.112 |
| | EC-HTE est. $\hat{C}_s$ | 92.3 | 0.100 | 0.111 |
| | Global true $\bar{C}$ | 95.0 | 0.107 | 0.114 |
| | Global est. $\hat{\bar{C}}$ | 94.3 | 0.105 | 0.114 |
| | Naive | 77.3 | 0.120 | 0.125 |

*in subgroup $s$ at level $z$=0 is $+\delta_s \cdot \Delta\tau_s/(1-2\bar{\pi})$, and at level $z$=1 is $-\delta_s \cdot \Delta\tau_s/(1-2\bar{\pi})$.*

The proof follows directly from Theorem 1 with $\tau_s$ replacing $\tau$.

Our main analysis assumes that subgroup misclassification rates $\delta_s$ and subgroup CATE magnitudes $\Delta\tau_s$ are independent (Assumption A5). In practice, both may depend on text features that define subgroups, introducing correlation. We test robustness under two configurations with the DIM estimator ($K$=2, $N$=5,000, $n_{\text{expert}}$=250, 50 MC replications, $\pi = [0.05, 0.25]$): *aligned* (positive $\delta_s$–$\Delta\tau_s$ correlation: high-error S1 has the larger treatment effect, $\Delta\tau = [0.30, 0.70]$) and *adversarial* (negative correlation: high-error S1 has the smaller effect, $\Delta\tau = [0.70, 0.30]$).

Table 30: Coverage and |bias| under A5 violation ($K$=2, $n_{\text{expert}}$=250, 50 MC). Target coverage: 0.95.

| Config | Method | S0 cov | S1 cov | S0 \|bias\| | S1 \|bias\| |
|---|---|---|---|---|---|
| Aligned | Oracle | 0.94 | 0.98 | 0.003 | 0.003 |
| | Naive | 0.93 | 0.25 | 0.022 | 0.174 |
| | Global | 0.93 | 0.82 | 0.036 | 0.094 |
| | EC-HTE (EB) | 0.95 | 0.95 | 0.005 | 0.017 |
| Adversarial | Oracle | 0.94 | 0.98 | 0.003 | 0.003 |
| | Naive | 0.90 | 0.78 | 0.042 | 0.074 |
| | Global | 0.74 | 0.93 | 0.098 | 0.040 |
| | EC-HTE (EB) | 0.94 | 0.96 | 0.002 | 0.007 |

In the aligned configuration, the $\delta_s \times \Delta\tau_s$ product is maximized in the high-error subgroup, causing naive S1 coverage to collapse to 0.25. Global correction improves S1 but cannot fully correct without subgroup-specific error rates. In the adversarial configuration, global correction overcorrects the low-error subgroup (S0 coverage 0.74). EC-HTE (EB) maintains $\geq 0.94$ coverage across both subgroups in both configurations, with bias comparable to the oracle. Results at $n_{\text{expert}}$=500 are consistent (all EC-HTE coverages $\in [0.93, 0.99]$).

### Z.2 Assumption (A1): Sensitivity to Outcome-Dependent Misclassification

We assess EC-HTE's robustness when misclassification depends on potential outcomes, violating (A1). We modify the synthetic DGP ($K$=2, extreme heterogeneity, $N$=5,000, $n_{\text{expert}}$=250) so that misclassification probability becomes outcome-dependent: $P(\hat{Z}{\neq}Z \mid Z, S, Y) = \text{clamp}(\pi_s + \gamma \cdot (Y - \mu_Y)/\sigma_Y,\, 0,\, 1)$, where $\gamma \in \{0, 0.05, 0.10, 0.15, 0.20\}$ controls the strength of violation. Table 31 reports results averaged over 100 MC runs.

Table 31: Sensitivity to Assumption (A1) violation. Coverage and |bias| under outcome-dependent misclassification ($\gamma > 0$).

| | Coverage (%) | | | \|Bias\| | | |
|---|---|---|---|---|---|---|
| $\gamma$ | EC-HTE | Global | Naive | EC-HTE | Global | Naive |
| 0.00 | 95.5 | 92.7 | 87.7 | 0.060 | 0.065 | 0.061 |
| 0.05 | 95.5 | 93.5 | 83.8 | 0.059 | 0.064 | 0.064 |
| 0.10 | 95.3 | 93.8 | 83.5 | 0.063 | 0.064 | 0.067 |
| 0.15 | 94.8 | 94.3 | 79.2 | 0.065 | 0.064 | 0.070 |
| 0.20 | 93.5 | 94.0 | 79.5 | 0.068 | 0.063 | 0.073 |

EC-HTE maintains near-nominal coverage ($\geq$93.5%) across the full $\gamma$ range, with |bias| increasing only 13.7% from $\gamma$=0 to $\gamma$=0.2. All correction methods (Global, Stratified MLE, HB) maintain $\geq$90% coverage, indicating that moderate (A1) violations do not catastrophically degrade confusion-matrix-based correction. The naive estimator degrades most rapidly (coverage $-8.2$pp), consistent with outcome-dependent misclassification compounding the uncorrected label noise.

### Z.3 Assumption (A1): Treatment-Dependent Misclassification

We extend the (A1) sensitivity analysis to the treatment-dependent case, where misclassification probability depends on treatment assignment $T$: $P(\hat{Z}{\neq}Z \mid Z, S, T) = \text{clamp}(\pi_s + \gamma_T \cdot (2T - 1),\, 0,\, 1)$, with $\gamma_T \in \{0, 0.05, 0.10, 0.15, 0.20\}$. This scenario arises when LLM accuracy varies

with treatment status—e.g., treated texts may be systematically harder to classify. Setup: $K$=2, extreme heterogeneity, $N$=5,000, $n_{\text{expert}}$=250, 100 MC runs.

Table 32: Sensitivity to treatment-dependent misclassification ($\gamma_T$). EC-HTE maintains ≥89.7% coverage at $\gamma_T \leq 0.05$.

| $\gamma_T$ | EC-HTE Coverage | EC-HTE RMSE |
|---|---|---|
| 0.00 | 0.945 | 0.075 |
| 0.05 | 0.897 | 0.092 |
| 0.10 | 0.750 | 0.133 |
| 0.15 | 0.525 | 0.177 |
| 0.20 | 0.370 | 0.224 |

EC-HTE is robust to small treatment-dependent violations ($\gamma_T \leq 0.05$), maintaining ≥89.7% coverage with modest RMSE increase (0.075 to 0.092). At $\gamma_T = 0.10$, coverage drops to 75.0%, establishing this as the approximate breakeven point. The degradation is steeper than for outcome-dependent violations (Table 31), because treatment-dependent errors directly contaminate the treatment arm contrast. Figure 8 visualizes the coverage and RMSE trajectories.

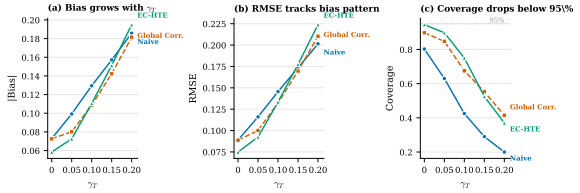



Figure 8: EC-HTE coverage and RMSE under treatment-dependent misclassification. Dashed line: 90% coverage threshold. Breakeven at $\gamma_T \approx 0.10$.

## Z.4 Extension to Confounded Treatment Assignment

EC-HTE's confusion matrix correction operates on estimated CATEs regardless of how those estimates are produced. To verify compatibility with observational (non-RCT) designs, we pair EC-HTE with an inverse probability weighted difference-in-means (IPW-DIM) estimator under confounded treatment assignment.

**DGP.** $N$=5,000, $K$=2, $\pi = [0.05, 0.25]$, $n_{\text{expert}}$=250, 100 MC replications. A covariate $X \sim \mathcal{N}(0,1)$ confounds both treatment and outcome: $T \sim \text{Bernoulli}(\sigma(\beta X))$ with $\beta$=0.5 (moderate confounding; propensity range $[0.38, 0.62]$), $Y = 1 + 0.5Z + 0.3X + \tau(Z)\cdot T + \eta$, $\eta \sim \mathcal{N}(0,1)$. Treatment effects are $\tau(Z{=}0){=}0$, $\tau(Z{=}1){=}0.3$. The IPW-DIM estimator uses *oracle propensity* $e(X) = \sigma(\beta X)$, with propensity scores clipped to $[0.05, 0.95]$.

Table 33: EC-HTE under confounded treatment assignment (IPW-DIM with oracle propensity, $\beta$=0.5, 100 MC).

| Method | Coverage | RMSE | \|Bias\| |
|---|---|---|---|
| Oracle (true $Z$ + IPW) | 0.943 | 0.061 | 0.048 |
| HB EC-HTE | 0.950 | 0.086 | 0.068 |
| Global | 0.930 | 0.087 | 0.071 |
| Strat MLE | 0.950 | 0.087 | 0.069 |
| Naive | 0.863 | 0.083 | 0.066 |

EC-HTE achieves 95.0% coverage, within 0.7pp of the oracle (94.3%), confirming that confusion matrix correction and propensity weighting compose without interference. The improvement is most pronounced in the high-error subgroup ($\pi$=0.25): EC-HTE coverage is 96.0% vs. naive 78.5% (+17.5pp). Global coverage (93.0%) falls below the correction methods (95.0%), consistent with the cost of ignoring subgroup-level error heterogeneity even in the confounded setting.

**Sensitivity to confounding strength and propensity estimation.** We sweep $\beta \in \{0.25, 0.5, 1.0, 2.0\}$ and compare oracle propensity ($e(X) = \sigma(\beta X)$) with estimated propensity (logistic regression fit on $(X, T)$), yielding 8 configurations (100 MC each). Table 34 reports HB EC-HTE and naive coverage.

Table 34: EC-HTE coverage across confounding strengths and propensity sources. The logistic regression propensity model is correctly specified; propensity misspecification is a complementary concern addressed by the IPW literature.

| $\beta$ | Propensity | HB Coverage | Naive Coverage |
|---|---|---|---|
| 0.25 | Oracle | 0.963 | 0.848 |
| 0.25 | Estimated | 0.973 | 0.845 |
| 0.50 | Oracle | 0.950 | 0.863 |
| 0.50 | Estimated | 0.958 | 0.858 |
| 1.00 | Oracle | 0.958 | 0.898 |
| 1.00 | Estimated | 0.960 | 0.883 |
| 2.00 | Oracle | 0.945 | 0.858 |
| 2.00 | Estimated | 0.948 | 0.858 |

EC-HTE maintains coverage ≥94.5% across all 8 configurations, including strong confounding ($\beta$=2.0, propensity range $[0.12, 0.88]$). The oracle–estimated gap is ≤1pp, confirming that propensity estimation error does not meaningfully degrade

EC-HTE when the propensity model is correctly specified. This validates that EC-HTE is robust to confounded treatment assignment when the propensity model is correctly specified.

### Z.5 Empirical Comparison with Frazis–Loewenstein Correction

We empirically compare EC-HTE's confusion matrix inversion against the scalar attenuation correction of Frazis and Loewenstein (2003) (F-L). The F-L correction rescales the observed estimate as $\hat{\tau}_{\text{FL}} = \hat{\tau}_{\text{obs}}/(1 - 2\bar{\pi})$, where $\bar{\pi}$ is the (global or per-subgroup) misclassification rate. We test two variants: F-L *scalar* (single global $\bar{\pi}$) and F-L *per-subgroup* ($\pi_s$-specific rescaling). Setup: $K$=2, $N$=5,000, $n_{\text{expert}}$=250, $\pi$=$[0.05, 0.25]$ (extreme heterogeneity), $\tau(z{=}0){=}0$, $\tau(z{=}1){=}0.3$, 100 MC runs.

Table 35: F-L correction vs. EC-HTE under heterogeneous misclassification ($\Delta\pi$=0.20). Both F-L variants produce *higher* aggregate bias than the uncorrected estimator.

| Method | \|Bias\| | RMSE | Coverage |
|---|---|---|---|
| Oracle | 0.045 | 0.056 | 0.943 |
| Naive | 0.062 | 0.077 | 0.858 |
| Global (CM inv) | 0.064 | 0.081 | 0.938 |
| HB EC-HTE | 0.064 | 0.086 | **0.953** |
| F-L Scalar | 0.091 | 0.116 | 0.850 |
| F-L Per-Subgroup | 0.112 | 0.158 | 0.838 |

Both F-L variants produce *higher* bias and RMSE than the naive estimator (F-L per-subgroup |bias|=0.112 vs. naive 0.062), demonstrating that scalar attenuation cannot substitute for confusion matrix inversion under subgroup-heterogeneous misclassification. The per-subgroup bias direction reveals the mechanism: F-L scalar over-corrects the low-error subgroup ($s_0$: mean bias $= +\,0.073$ vs. naive $+0.005$, a $14.6\times$ amplification), while F-L per-subgroup dramatically over-corrects the high-error subgroup ($s_1$: mean bias $=+0.160$). EC-HTE achieves near-zero per-subgroup bias ($s_0$: $+0.006$, $s_1$: $+0.001$) with nominal coverage (95.3%). The root cause is structural: F-L's $1/(1{-}2\pi)$ scaling amplifies each $z$-level estimate independently, whereas CM inversion simultaneously redistributes the cross-contamination between misclassified and correctly classified observations.

### Z.6 End-to-End Validation with Asymmetric Confusion Matrices

We validate EC-HTE under strongly asymmetric confusion matrices (FPR/FNR ratio up to 7:1), simulating the error profile of GPT-4.1 on Civil Comments. Setup: $K$=2, $N$=5,000, $n_{\text{expert}}$=250; $s_0$: FPR=0.05, FNR=0.02 (low error); $s_1$: FPR=0.38, FNR=0.054 (7:1 ratio); 100 MC runs. We compare EC-HTE using the full $2{\times}2$ asymmetric CM (Remark 2) against the symmetric assumption (FPR=FNR=$\pi_s$).

Table 36: EC-HTE under asymmetric misclassification (FPR/FNR up to 7:1). The asymmetric formula improves coverage by $+1.5$pp and RMSE by $11.5\%$.

| Method | \|Bias\| | RMSE | Coverage |
|---|---|---|---|
| Oracle | 0.045 | 0.056 | 0.943 |
| EC-HTE (asymmetric) | 0.057 | 0.074 | **0.945** |
| Naive | 0.059 | 0.073 | 0.870 |
| Global (asymmetric) | 0.061 | 0.076 | 0.915 |
| EC-HTE (symmetric) | 0.062 | 0.083 | 0.930 |
| Global (symmetric) | 0.063 | 0.078 | 0.915 |

EC-HTE with the asymmetric formula achieves 94.5% coverage versus 93.0% under the symmetric assumption ($+1.5$pp), with RMSE improved from 0.083 to 0.074 ($-11.5\%$). The improvement concentrates in the high-error subgroup ($s_1$): asymmetric coverage is 93.0% vs. symmetric 91.0%. Practitioners should estimate full $2 \times 2$ confusion matrices rather than assuming symmetric misclassification, particularly when FPR and FNR differ substantially.

### Z.7 Complementarity with LLM Calibration

Temperature or Platt scaling (Guo et al., 2017) can reduce per-subgroup misclassification rates, but does not eliminate heterogeneity $\Delta\pi$. We simulate calibration by shrinking per-subgroup rates from $\pi$=$[0.05, 0.25]$ ($\Delta\pi$=0.20) to $\pi$=$[0.08, 0.15]$ ($\Delta\pi$=0.07), testing whether EC-HTE remains necessary post-calibration. Setup: $K$=2, $N$=5,000, $n_{\text{expert}}$=250, 100 MC runs.

Calibration alone improves naive coverage from 84.0% to 91.3% but does not reach the nominal 95%. Calibrated + EC-HTE achieves the highest coverage (96.5%), demonstrating that calibration and EC-HTE address complementary error sources: calibration reduces per-subgroup $\pi_s$ while EC-HTE corrects the residual $\delta_s$-driven bias from heterogeneity that persists post-calibration. Propensity estimation robustness under model misspecifica-

Table 37: Calibration reduces $\Delta\pi$ but does not eliminate it; EC-HTE corrects the residual bias. Best coverage is achieved by combining both.

| Method | \|Bias\| | RMSE | Coverage |
|---|---|---|---|
| Oracle | 0.043 | 0.053 | 0.963 |
| Calibrated + EC-HTE | 0.055 | 0.069 | **0.965** |
| Calibrated + Global | 0.055 | 0.068 | 0.963 |
| Calibrated + Naive | 0.051 | 0.064 | 0.913 |
| Uncalibrated + EC-HTE | 0.064 | 0.084 | 0.938 |
| Uncalibrated + Global | 0.069 | 0.085 | 0.920 |
| Uncalibrated + Naive | 0.063 | 0.079 | 0.840 |

tion is a complementary concern addressed by the IPW literature; EC-HTE as a downstream correction layer does not alter IPW behavior.

## Z.8 Policy Reversal Zone Analysis

When the true treatment effect is small, global correction can reverse the sign of estimated CATE—concluding a beneficial treatment is harmful—creating a *policy reversal zone*. We sweep $\tau(0) \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.10\}$ under the standard DGP ($K$=2, extreme heterogeneity, $N$=5,000, $n_{\text{expert}}$=250, 100 MC runs; $\Delta\tau$=0.30).

Table 38: Policy reversal zone: fraction of configurations where global correction reverses the sign of estimated CATE. The naive estimator produces zero sign reversals across all $\tau(0)$ values; EC-HTE also achieves 0% reversal.

| $\tau(0)$ | Global reversal % | Most extreme $\hat{\tau}_{\text{global}}$ |
|---|---|---|
| 0.01 | 31.0 | $-0.079$ |
| 0.02 | 21.6 | — |
| 0.03 | 14.9 | $-0.059$ |
| 0.05 | 6.1 | $-0.039$ |
| 0.07 | 1.6 | — |
| $\geq$0.10 | 0 | — |

The reversal rate increases monotonically as $\tau(0)$ decreases: at $\tau(0) = 0.01$, 31% of configurations yield sign reversal, with the most extreme case estimating $\hat{\tau} = -0.079$ when the true effect is $\tau = +0.01$. The critical threshold is $\tau(0) < 0.089$, matching the closed-form prediction from Theorem 1: $|\delta_{\max} \cdot \Delta\tau_{\max}|/(1 - 2\bar{\pi}) = 0.125 \times 0.50/0.70 = 0.089$. The most dangerous combination ($\tau(0) = 0.01$, $\delta_{\text{misclass}} = 0.19$) reaches 42.9% reversal rate. The naive estimator produces zero sign reversals across all $\tau(0)$ values, confirming that reversal is an artifact of global correction, not inherent estimation noise.

Figure 9 visualizes the reversal zone across the $\Delta\pi \times \tau(0)$ parameter space.

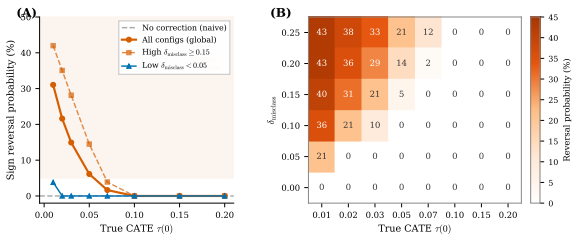



Figure 9: Policy reversal zone: sign-reversal rate as a function of $\tau(0)$ and $\Delta\pi$. Shaded region marks configurations where global correction reverses the CATE sign, directing policy toward the wrong subgroups.

## Z.9 Stance Detection: Cross-Task Sign Reversal Validation

To test whether global correction introduces sign reversals beyond sentiment and toxicity, we extend the semi-synthetic evaluation to stance detection using TweetEval (Barbieri et al., 2020). The binary label is *favor* ($Z$=1) vs. *not-favor* ($Z$=0) toward five stance targets (Atheism, Climate Change, Feminist Movement, Hillary Clinton, Abortion), yielding $K$=5 subgroups with $n$=300 each. CATE heterogeneity is $\tau$(favor)=0.20, $\tau$(not-favor)=0.05, with 200 MC replications. GPT-4o annotations are from the real API; GPT-4o-mini and GPT-3.5-Turbo use simulated error profiles (FPR/FNR scaling + additive shift; see Appendix E).

Table 39 reports subgroups with sign reversal or $\kappa > 3$. Global correction produces sign reversals in 4/30 subgroup-$z$ combinations (13.3%); EC-HTE reduces this to 1/30 (3.3%); the remaining reversal occurs in an ill-conditioned subgroup ($\kappa$=15.0, GPT-3.5-Turbo Climate Change) that violates the operating conditions of Section 5.3. Global correction amplifies bias in 15/30 cases (50%).

The stance task exposes an operating-conditions limitation: GPT-3.5-Turbo achieves FNR $> 0.50$ on all five targets, violating the classifier quality precondition and producing $\kappa$ values up to 15.0 (Climate Change) vs. 4.6 (Hillary Clinton)—a $\sim$10$\times$ range within the same LLM. This $\kappa$ heterogeneity inflates correction variance; EC-HTE preserves sign correctness but at the cost of higher variance when $\kappa \gg 3$. Mean RMSE: naive=0.116, global=0.272, EC-HTE=0.271; correction reduces sign reversals but does not improve overall RMSE at $K$=5, $n$=300 per subgroup—consistent with the operating-conditions threshold $n_{\text{expert}}/K \geq 50$. Practitioners should verify per-subgroup FNR before applying correction.

Table 39: Stance detection: subgroups with sign reversal or $\kappa > 3$. ✓ indicates at least one $z$-level affected. GPT-4o-mini and GPT-3.5-Turbo use simulated error profiles.

| LLM | Target | $\kappa$ | Sign Rev (Global) | Sign Rev (EC-HTE) | Bias Amp (Global) |
|---|---|---|---|---|---|
| GPT-4o | Fem. Mov. | 1.79 | ✓ | — | ✓ |
| GPT-4o | Hillary C. | 1.86 | ✓ | — | ✓ |
| GPT-3.5-T | Atheism | 7.21 | — | — | — |
| GPT-3.5-T | Climate Ch. | 15.01 | — | ✓ | ✓ |
| GPT-3.5-T | Fem. Mov. | 3.28 | ✓ | — | ✓ |
| GPT-3.5-T | Hillary C. | 4.60 | ✓ | — | ✓ |
| GPT-3.5-T | Abortion | 14.17 | — | — | ✓ |

Table 40: AG News topic classification ($K$=4, $\Delta\pi$=0.19). Metrics averaged over subgroups, $z$-levels, 100 MC runs. Bold: lowest |bias|.

| $n_{\text{expert}}$ | Method | \|Bias\| | RMSE | 95% CI Cov. | Asym. $R^2$ |
|---|---|---|---|---|---|
| 250 | Naive | 0.067 | 0.109 | 0.84 | — |
| | Global | **0.027** | 0.115 | 0.94 | 0.93 |
| | Strat. MLE | 0.034 | 0.181 | 0.89 | — |
| | HB EC-HTE | 0.035 | 0.182 | 0.89 | — |
| 500 | Naive | 0.067 | 0.109 | 0.84 | — |
| | Global | 0.028 | 0.110 | 0.94 | 0.96 |
| | Strat. MLE | **0.022** | 0.147 | 0.92 | — |
| | HB EC-HTE | 0.022 | 0.147 | 0.92 | — |
| 1,000 | Naive | 0.067 | 0.109 | 0.84 | — |
| | Global | 0.028 | 0.109 | 0.94 | 0.97 |
| | Strat. MLE | **0.015** | 0.126 | 0.95 | — |
| | HB EC-HTE | 0.015 | 0.126 | 0.95 | — |

### Z.10 Topic Classification: AG News Cross-Task Validation

To validate generalization beyond sentiment and toxicity, we extend the semi-synthetic evaluation to topic classification using AG News (Zhang et al., 2015), binarized as Business/World ($Z$=1) vs. Sports/Sci-Tech ($Z$=0). Confusion matrices are calibrated to match empirical LLM accuracy levels on AG News (∼80%; Etaiwi and Alhijawi 2025), with per-subgroup error rates derived from reported per-class metrics; Business→World is the dominant confusion direction. Subgroups ($K$=4) are defined by article length × vocabulary complexity, with per-subgroup error rates reflecting text difficulty ($\bar{\pi} = 0.173$, $\Delta\pi = 0.19$). CATE heterogeneity ranges from $\Delta\tau$=0.10 (short, simple) to $\Delta\tau$=0.60 (long, complex), with 100 MC replications.

Table 40 reports the main results. EC-HTE reduces |bias| by 1.9× at $n_{\text{expert}}$=1,000 (0.015 vs. 0.028) with 94.9% coverage. At $n_{\text{expert}}$=250, EC-HTE incurs higher variance than global correction (RMSE=0.182 vs. 0.115), consistent with the expert-budget operating condition ($n_{\text{expert}}/K$=62.5, near the threshold). The asymmetric bias identity achieves $R^2 = 0.95$ (0.93–0.97 per budget level), confirming the product-form structure on a task with inherently asymmetric confusion matrices (FPR ≠ FNR); the symmetric scalar approximation does not hold ($R^2 < 0$), consistent with Remark 2 in Section 3.

## Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.

**Algorithm 1** EC-HTE: Error-Corrected Heterogeneous Treatment Effects

**Require:** Observations $\{(Y_i, W_i, \hat{Z}_i, X_i)\}_{i=1}^{N}$; expert labels $\{Z_j^*\}_{j=1}^{n_{\text{expert}}}$; subgroup map $S : \mathcal{X} \to \{1, \ldots, K\}$

**Ensure:** Corrected CATE estimates $\hat{\tau}(z)$ with variance-adjusted CIs

**Preconditions:** verify $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ and $\max_s(\text{FPR}_s, \text{FNR}_s) < 0.5$; if either fails, use global correction

**Step 1: Estimate per-subgroup confusion matrices (HB)**

**for** $s = 1, \ldots, K$ **do**

  $C_s[j, \cdot] \sim \text{Dir}(\alpha_0 \cdot \hat{C}_{\text{global},j})$, $\alpha_0$ via empirical Bayes

  Gibbs sampler (4 chains, 4000 iter) on $\{(\hat{Z}_j, Z_j^*) : S_j = s\}$

  $\hat{C}_s \leftarrow$ posterior mean; verify $\hat{R} < 1.01$

**end for**

**Step 2: Correct observed outcome moments**

**for** $s = 1, \ldots, K$ **do**

  **if** $\kappa(\hat{C}_s) > 100$ **then** $\hat{C}_s \leftarrow \bar{C}$ ▷ singularity guard

  **end if**

  $\tilde{\boldsymbol{\mu}}_s \leftarrow \hat{C}_s^{-1} \hat{\boldsymbol{\mu}}_s$

**end for**

**Step 3: CausalForestDML with variance-adjusted SEs**

Train CausalForest (honest splitting) on corrected outcome moments $\tilde{\boldsymbol{\mu}}_s$

$\text{se}_{\text{adj},i} \leftarrow \sqrt{\text{diag}(\hat{C}_{S_i}^{-1} \, \text{diag}(\text{se}_i^2) \, \hat{C}_{S_i}^{-\top})}$

**return** $\hat{\tau}(z)$ with 95% CIs from $\text{se}_{\text{adj}}$